



**Verified Carbon  
Standard**

# VALIDATION REPORT OF IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Document Prepared by Earthood Services Private Limited

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| <b>Project Title</b> | Improved Cookstove Grouped Project in Papua New Guinea |
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| <b>Report Title</b>  | Validation Report of Improved Cookstove Grouped Project in Papua New Guinea                                                                                                                                                                                                |  |  |
| <b>Client</b>        | Tasman Environmental Markets Asia Pacific                                                                                                                                                                                                                                  |  |  |
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|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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## Summary:

Earthood Services Private Limited (hereafter referred to as Earthood) has been contracted by Tasman Environmental Markets Asia Pacific to conduct the validation report of the grouped project – “Improved Cookstove Grouped Project in Papua New Guinea”, VCS ID: 4416 regarding the relevant requirements of VCS programme guidelines and standard (VCS Standard Version 4.5, & VCS Program Guide Version 4.4)/6//7/.

The project is a grouped project that uses the VCS methodology, VMR0006 version 1.2 /04/, which mandates the use of AMS-II.G. version 13.0/4/. The project involves the distribution of fuel efficient improved cookstoves (ICS) within the boundary of Papua New Guinea (PNG). The ICS disseminated through this project will replace open fired wood fueled cooking methods.

The objective of the validation is to perform a complete and independent review of the proposed project activity against the applicable VCS requirements, including the baseline, monitoring plan, and compliance with the relevant VCS and host country criteria. These are validated to ensure that the project design, as documented, is sound and reasonable, and that it fits the established standards. Validation is a prerequisite for all the VCS projects, and it is viewed as vital to give stakeholders assurance about the project’s quality and expected emission reductions. The VVB’s aim is to perform a complete, independent evaluation of the project activity.

The scope of validation is described as an objective and independent review of the project description (PD). The PD is compared to the relevant criteria and guidance documents provided by VCS, including the VCS Program Guide/7/, VCS Standard/6/, Program Definitions/15/, and the Registration & Issuance Process in order to ensure that the project meets the applicability conditions of the chosen baseline and monitoring methodology, VMR0006 version 1.2 /04/.

The validation, which was conducted following Earthood’s internal quality procedures, consisted of the following phases:

- a) Document review, involving:
  - A review of data and information.
  - Cross checks between information provided in the PD and information from sources other than those used, if available, and if necessary independent background investigations.
- b) Follow-up actions (e.g., site visit and telephone or email interviews), including:
  - Interviews with relevant stakeholders in the host country, personnel with knowledge of the project design and implementation.
  - Cross checks between information provided by interviewed personnel (i.e., by checking sources or other interviews) to ensure that no relevant information has been omitted.

- c) Reference to available information relating to projects or technologies similar to the proposed project activity under validation.
- d) Review, based on the approved methodology being applied, of the appropriateness of formulae and accuracy of calculations.

A risk-based approach has been followed to perform this validation. During the validation process, a total of 16 findings were raised, which includes 05 Corrective Action Requests (CARs), 11 Clarification Requests (CLs) and 01 Forward Action Requests (FARs). The project proponent has successfully closed all the issues raised.

The PD, emissions reduction calculations, and accompanying documents provided are all in compliance with VCS criteria. The validation was completed with a reasonable level of assurance and no uncertainties were found related to the project activity validation.

Earthood Services Private Limited (hereafter referred to as Earthood) certifies that the project meets the requirements of the VCS PD template Project Description, v4.2, the VCS Standard, v4.5/6/ and VMR0006 v1.2 /04/. The estimated annual average and total emission reductions from the project activity, for the crediting period 01/01/2024 to 31/12/2033 are expected to be 242,804 tCO<sub>2e</sub> average annually and 2,428,045 tCO<sub>2e</sub> over the crediting period and has thus been successfully validated under VCS. Earthood issues a positive validation opinion, certifying that the project meets the applicable VCS standards and thereby proposing that it be registered.

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# 1 INTRODUCTION

## 1.1 Objective

Earthood Services Private Limited (hereafter referred as Earthood) has been contracted by Tasman Environmental Markets Asia Pacific to conduct the validation of the VCS grouped project “Improved Cookstove Grouped Project in Papua New Guinea” (VCS ID 4416) located in Papua New Guinea (PNG) to carry out validation of the project against the relevant requirements of VCS programme guidelines and standard (VCS Standard Version 4.5/6/, & VCS Program Guide Version 4.4/7/).

The purpose of this validation is to have an independent third-party assessment of whether the project activity conforms to the qualification criteria set out in the VCS Standard, version 4.5/6/ to attain real, measurable, additional, and permanent emission reductions. The validation statement opinion is a written assurance that:

- The project complies with all the applicable VCS requirements and can generate the emission reductions stated over the project’s crediting period.
- The validation followed the requirements of the current version of the VCS Standard, version 4.5/6/ and VCS Program Guide, version 4.4/7/ to ensure the quality and consistency of the validation work and the report.
- The project shall result in emission reductions as declared by the organisation or GHG project’s GHG assertion.

In order to fulfil above objectives, the VVB assigned the task of assessment to validation team that is collectively qualified as per Earthood’s QMS manual and other internal requirements. The data reported is accurate, complete, consistent, transparent, and free of material error or omission.

## 1.2 Scope and Criteria

The scope of the validation is carried out in to establish/verify that:

- The project activity is in accordance with rules and legislation of the host country (Papua New Guinea).
- The project activity is in accordance with all relevant VCS rules and requirements, which includes the following: VCS Program Guide v.4.4/7/ and VCS Standard v.4.5/6/, Program Definitions v.4.4/15/, Registration and Issuance Process v.4.3/32/ and VCS Validation and Verification Manual v.3.2/8/, including additionality, proof of title and compliance with local laws.
- The project activity is in accordance with the conditions of the latest version of applied VCS methodology, VMR0006 v1.2 /04/.

The validation is based on the data provided to Earthood as well as the contract terms. The validation is not intended to provide any consulting towards the Client. However, indicated requests for forward and/or corrective actions may provide information for improving project monitoring to reduce GHG emissions.

The validation process is divided into four stages:

1. Desk review of Project Description (PD)
  - Evaluation of data and information
  - Cross-checks between information provided in the PD/1/ and information obtained from other sources using all available resources without regard to the project proponent's information.
2. On-site/ Remote site visit and follow-up interviews with project stakeholders
  - Telephone, email, or direct on-site interviews with relevant stakeholders in the host country with professionals with expertise of the project development.
  - Cross-checking the information provided by interviewed people with the information provided by the project proponent using all available means.
3. Reference to publicly available information about projects or technologies that are similar to the project under validation and review, using the accepted methodology for formulae and calculation accuracy.
4. The final validation report and opinion, as well as the resolution of remaining concerns.

### 1.3 Reasonableness of Assumptions

☐ Limited Level of Assurance

☒ Reasonable Level of Assurance

The approach used by Earthood for validation of the crediting period is built on a thorough understanding of the risks associated with reporting data on GHG emissions and the controls used to mitigate them. Earthood conducted the validation by reviewing substantiating evidence and other relevant information and explanations from sources to provide reasonable assurance that estimated GHG emission reductions are fairly reported.

In the draft validation report (prepared by assessment team), the information provided is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures established and implemented by Earthood were duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable VCS and CDM (Clean Development Mechanism) requirements as appropriate. The technical review team is collectively required to possess technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of the technical review team are independent of the validation team. The report approved by the Technical Manager is endorsed by the Managing Director, who is overall responsible for ensuring quality, before final release. Further details of applicable procedures and responsibilities concerning the Earthood Quality Management System (QMS) are available on its website ([www.earthood.in](http://www.earthood.in)).

In our opinion the estimated and actual GHG emissions reductions were calculated correctly based on the approved baseline and monitoring methodologies VCS methodology, VMR0006 v1.2\04/. The assessment result has a reasonable level of assurance in validation that GHG assertions are free of material errors, omissions, and misrepresentations. The document reviewed are included under Appendix 1 of this report.

## 1.4 Summary Description of the Project

The grouped project activity 'Improved Cookstove Grouped Project in Papua New Guinea' involves the distribution of fuel efficient improved cookstoves (ICS) within the boundary of Papua New Guinea (PNG). The project aims to replace the open-fired wood-fuelled cooking methods by disseminating these ICS. In phase I of the project, a total of 40,000 ICS is planned to be distributed, which will eventually scale up to 100,000 ICS. G3 Burn rocket stove (also known as Kuniokoa or Ecoa wood cookstove) with an adjustable pot skirt is being distributed to each household. One project device will be provided to each household coming under the project activity. These high efficiency stoves replace the existing inefficient, conventional open fire stoves. These improved cookstoves will burn more efficiently as compared to the baseline stoves and thus will save the amount of wood burnt in cooking purpose and emissions associated with it. The project also reduces deforestation in host country and minimizes the health risks associated with indoor smoke pollution. The project further also reduces the amount of time spent by end-users foraging for firewood.

The validation team has verified that the technology distributed under the group project activity is G3 Burn rocket stove (also known as Kuniokoa or Ecoa wood cookstove) with an adjustable pot skirt with technical specifications/9/ as following:

|                                   |                                  |
|-----------------------------------|----------------------------------|
| Thermal efficiency with pot skirt | 50.8%                            |
| Dimensions                        | 32.1 cm height, 28.2 cm diameter |
| Lifespan                          | 10 Years                         |

The efficiency and technical specifications of the cookstoves is verified by the WBT test report of CREEC/14/ for G3 Burn rocket stove. The assessment team has verified the test report submitted, it has been verified that efficiency is calculated in accordance with the applied methodology, VMR0006 v1.2 /4/.

The Tasman Environmental Markets Asia-Pacific Pte Ltd is the project proponent and also owns the rights to VERs, Declaration/12.2/ for the same has been submitted by the PP to the assessment team. It has also been verified that individual households using ICS will sign a contract with PP, under which the end users will certify the use of ICS and transfer the VCSs right to project proponent/30/.

The project is a grouped project with estimated 242,804 tCO<sub>2</sub>e of emission reductions annually and 2,428,045 tCO<sub>2</sub>e of emission reduction over the entire crediting period i.e., 01/01/2024 to 31/12/2033. The start of the group project activity is 01/01/2024, which is also the estimated date of commissioning of ICS Phase I under the group project activity/10/.

## 2 VALIDATION PROCESS

## 2.1 Method and Criteria

Tasman Environmental Markets Asia-Pacific Pte Ltd have contracted Earthood, to carry out the validation of the grouped project “Improved Cookstove Grouped Project in Papua New Guinea”, with regards to the relevant requirements of VCS Standard, Version 4.5/6/. The project activity uses the VMR0006. version 1.2/4/, which is classified as sectoral scope 3 “Energy Demand”. The following were included in the validation:

Contract with PP for the scope and appointment of validation team and technical review team.

Completeness check of VCS-PD/2/

Desk review of VCS-PD/2/ and estimated GHG emission reduction calculation sheet/3/.

Remote site inspection by validation team (due to the prevailing health concerns in the host country).

Follow up interviews with PP and Project Owner

Reporting and closure of findings (CARs/CLs/FARs) and preparation of draft validation report

Independent technical review of the draft validation report and final/ revised documentation (e.g. VCS PD, corresponding estimated ER calculation sheet and evidence)

Issuance of the final validation report to the contracted.

## 2.2 Document Review

A desk review was conducted by the validation team that included:

- a. A review of the data and information presented to assess its completeness.
- b. A review of the VCS-PD/2/, estimated GHG emission reduction calculation sheet/3/, the applied methodology/4/ including applicable tool(s) and relevant laws regulations for the host country.
- c. Supporting documents as provided by the Project Participant.

In addition, the validation team consulted third-party documents such as host-party legislation, technical reports pertaining to the project design or design or basic condition, and technical data. A complete list of documents/evidence reviewed is included as Appendix 1.

## 2.3 Interviews

Though para 4.1.9 of the VCS standard v4.5 says that a site visit that includes a visit to facilities and/or project areas shall be conducted at validation, due to security concerns, the VVB has conducted a remote site visit on 22/09/2023 to check the baseline scenario and implementation plan of the project activity. A third-party security risk assessment by SecRisk has been conducted and has shared by PP/26/, through which The Letter of Exemption for the on-site visit has been received from Verra /33/. The remote visit was conducted as a part of validating the project to check the baseline technology used previously, the fuel used in baseline and project device and to check the



implementation of the improved stoves in the pilot project, along with the interviews of the LSC personnels and project proponent representatives. PP has conducted the baseline survey in 220 households across Western Highlands Province (WHP) and Southern Highlands Province (SHP) in Papua New Guinea (PNG) and has also installed a pilot cookstove project in these households for the pilot study of the cookstove. During the Remote Audit, the end-users were interviewed regarding the baseline technology they used, the current fuel type, the project technology they acquired, to check the additionality asked them if they have received the cookstove free of cost and so on. The following points were discussed as a part of the interviews:

- a) Introduction, scope and objective of work, roles and responsibilities, resources required, and timetable of the Remote audit including venue for closing meeting and any concerns from PP.
- b) All the details about the project activity mentioned in the PD was cross-checked by discussion with the PP.
- c) Conducting the remote audit interviewing the end-users
- d) And final submission of the audit findings to the client and agreement on the issues raised.

The table below describes the remote interview process and further identities personnel, including their roles, who were interviewed and/or provided information additional to that provided in the project description and any supporting document.

| S.No | Name                          | Organization                                       | Date       | Topic                                                                                   | Team                           |
|------|-------------------------------|----------------------------------------------------|------------|-----------------------------------------------------------------------------------------|--------------------------------|
| 1.   | Dr. Gabriel Machovsky Capuska | Tasman Environmental Markets Asia Pacific Pte. Ltd | 22/09/2023 | Project design, Project Implementation status, Project start date and Project Location, | Divij Varshney, Vaishali Yadav |
| 2.   | Cherie Sim                    | Tasman Environmental Markets Asia Pacific Pte. Ltd | 22/09/2023 | Baseline Scenario, Baseline Identification and                                          |                                |

|    |                 |                                                    |            |                                                                                                                                                                                                                                     |  |
|----|-----------------|----------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 3. | Michael Kisombo | Tasman Environmental Markets Asia Pacific Pte. Ltd | 22/09/2023 | Additionality, Qualification and Training, Monitoring and reporting documentation, Quality Assurance, Management and operating systems, local stakeholders meeting process, Compliance with relevant laws, Roles and responsibility |  |
|----|-----------------|----------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

#### Interview with the end users

| S.No. | Name           | Organization/<br>Affiliation               | Date       | Topic                                                                                                                                                                                                              | Team member    |
|-------|----------------|--------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.    | Donald Yaget   | End User- Western Highlands Province (WHP) | 22/09/2023 | Project stove Installation, Baseline stove and baseline fuel, Fuel used in project stove, Training and ongoing support by the PP for operation of the project stove, Project stove usage patterns, and Grievances. | Vaishali Yadav |
| 2.    | Betty Kaika    | End User- WHP                              | 22/09/2023 |                                                                                                                                                                                                                    | Vaishali Yadav |
| 3.    | Loise Kuli     | End User- WHP                              | 22/09/2023 |                                                                                                                                                                                                                    | Vaishali Yadav |
| 4.    | Sarah Yak      | End User- WHP                              | 22/09/2023 |                                                                                                                                                                                                                    | Vaishali Yadav |
| 5.    | Mekail         | End User- WHP                              | 22/09/2023 |                                                                                                                                                                                                                    | Vaishali Yadav |
| 6.    | Jonathan James | End User- WHP                              | 22/09/2023 |                                                                                                                                                                                                                    | Vaishali Yadav |
| 7.    | Jerman Okipul  | End User- Southern                         | 22/09/2023 |                                                                                                                                                                                                                    | Divij Varshney |

|     |                     |                          |            |  |                |
|-----|---------------------|--------------------------|------------|--|----------------|
|     |                     | Highlands Province (SHP) |            |  |                |
| 8.  | Clancy Sendil       | End User-SHP             | 22/09/2023 |  | Divij Varshney |
| 9.  | Yara Ungugomo       | End User-SHP             | 22/09/2023 |  | Divij Varshney |
| 10. | Ps Nelson Larsawan  | End User-SHP             | 22/09/2023 |  | Divij Varshney |
| 11. | Anna Kenneth Thomas | End User-SHP             | 22/09/2023 |  | Divij Varshney |
| 12. | Maria Joseph        | End User-SHP             | 22/09/2023 |  | Divij Varshney |

Additionally, to assess the capabilities of the ground staff to implement the project, 4 field assistants were remotely interviewed by the assessment team.

#### Interview with the Field Assistants

| S.No | Name           | Organization / Affiliation | Date       | Topic                                                                             | Team member                       |
|------|----------------|----------------------------|------------|-----------------------------------------------------------------------------------|-----------------------------------|
| 1.   | Arnold Terry   | Field Assistant            | 22/09/2023 | Trainings provided, baseline survey, implementation survey, registration process. | Divij Varshney and Vaishali Yadav |
| 2.   | Tony Yamba     | Field Assistant            | 22/09/2023 |                                                                                   | Divij Varshney and Vaishali Yadav |
| 3.   | Patrick Wapiyu | Field Assistant            | 22/09/2023 |                                                                                   | Divij Varshney and Vaishali Yadav |

|    |                     |                 |            |  |                                   |
|----|---------------------|-----------------|------------|--|-----------------------------------|
| 4. | Michael Kisomb<br>o | Field Assistant | 22/09/2023 |  | Divij Varshney and Vaishali Yadav |
|----|---------------------|-----------------|------------|--|-----------------------------------|

#### Interview with the Local Stakeholders Consultation (LSC) Meeting participants:

| S.No. | Name               | Organization/<br>Affiliation | Date       | Topic                                                                             | Team member    |
|-------|--------------------|------------------------------|------------|-----------------------------------------------------------------------------------|----------------|
| 1.    | Ps Enoc Wan        | LSC participant              | 22/09/2023 | Benefits of the project activity, Feedback about the project, Grievance register. | Vaishali Yadav |
| 2.    | George Herphit     | LSC participant              | 22/09/2023 |                                                                                   | Vaishali Yadav |
| 3.    | Janet Terry        | LSC participant              | 22/09/2023 |                                                                                   | Vaishali Yadav |
| 4.    | Magret Nip         | LSC participant              | 22/09/2023 |                                                                                   | Divij Varshney |
| 5.    | Peter Wapenda Wari | LSC participant              | 22/09/2023 |                                                                                   | Divij Varshney |
| 6.    | Joel Raus          | LSC participant              | 22/09/2023 |                                                                                   | Divij Varshney |

Through interviews and remote audit, the assessment team confirms the baseline scenario, baseline and project fuel type, additionality, local stakeholder consultation and its feedback. Thus, the assessment team achieved a reasonable level of assurance by conducting a remote site- visit of end-users.

## 2.4 Site Visits

Remote inspection was done for validation on 22/09/2023 to check the baseline technology used previously, the fuel used in baseline and project device and to check the implementation of the improved stoves. The assessment team has used remote audit interviews, survey etc. to support the validation criteria to reach a reasonable level of assurance. Earthood is able to confirm that the baseline scenario, baseline and project fuel type, additionality, and also, the pilot project is implemented in line with the description in the VCS PD/2/. Initially, the site visit was planned to be conducted physically, but due to security concerns in the region, the auditors conducted a remote audit on 22/09/2023 for the pilot project households, ground staff and LSC participants users. The team also interviewed the project proponent representatives on the same day. Therefore, assessment

team believes it achieved a reasonable level of assurance through various sessions and interviews with the PP, end-users and other stakeholders.

#### VVB Sampling Approach:

In line with para 28 of the CDM Sampling Standard/21/, “When the project participants or the coordinating/managing entity have applied a sampling approach, the DOE may apply acceptance sampling as described in the steps indicated in paragraphs 29–38 below as part of validation/verification activities” The validation team determined the sample size for acceptance sampling by evaluating the following, using its own professional judgement and guidance in the Standard ‘Sampling and surveys for CDM project activities and programme of activities, Version 9.0’:

- The proportion of discrepancies between the CME’s data and validation team’s (field or onsite inspection results) data that can be considered acceptable. This is referred to as the AQL (Acceptable Quality Level): 0.5% was considered in this assessment based on the VVB’s professional judgement.
- The proportion of discrepancies between the CME’s data and validation team’s (field or onsite inspection results) data that would be considered unacceptable. This is the UQL (Unacceptable Quality Level): 20% was considered in this assessment based on the VVB’s professional judgement.
- The producer risk and consumer risk of 10% and 10% respectively was considered as per the limit indicated in the para 31 of the sampling standard.

The Verification team selected the sample size as 11 project ICS consumers (as per Table 2 of CDM Sampling Standard) with 3 additional consumers as backup for the purpose of remote inspection to check the acceptability of PP’s sampling result or otherwise.

The verification team selected the random samples from PP’s baseline survey and pilot project sampled records to check the acceptability (or otherwise) of the data for each such record with PP’s baseline survey and pilot project sample records and determine if the PP’s sampling results is acceptable or otherwise.

Most of the households interviewed were found to have inefficient three stone stove as baseline stove using firewood which were replaced by the project stove in case of end-users. The ICSs disseminated had unique serial number and the end users confirmed that they have stopped using the three-stone fire stove and the cookstove was distributed free-of-cost to the households. A few households which received the ICS in the pilot-run admitted to own LPG and electric stoves as well, but their usage depended on the unreliable electricity and gas availability in the region. A finding CL#11 was raised in this regard. No other inconsistencies were observed. The end user details were also confirmed and found to be same as the information provided in the distribution database provided by the PP/23/.

## 2.5 Resolution of Findings

This section represents the conclusions from the validation of the project activity. The results of the document review, site visit/ remote evaluations, and interviews are presented in this section. CARs, CLs and FARs are used to correct material inconsistencies discovered during validation.

Corrective action requests (CARs) are issued where:

- i. Mistakes have been made with a direct influence on project results requiring adjustments of the VERs/ VCUs monitoring report.
- ii. Applicable methodological specific requirements have not been met.
- iii. There is a risk that emission reductions cannot be monitored or calculated.

A clarification Requests (CL) maybe used where additional information is needed to fully clarify an issue or where the information is not transparent enough to establish whether a requirement is met.

- i. The actual project monitoring and reporting practices requires attention and/ or adjustment for the next consecutive verification period, or
- ii. An adjustment of the monitoring plan/02/ is recommended.

In the context of FARs, risks have been identified, which may endanger the delivery of high-quality emissions reductions in the future, i.e., by deviations from standard procedures as defined by the Monitoring plan. Consequently, such aspects should receive a special focus during the next consecutive verification. A FAR may originate from lack of data sustaining claimed emission reductions. FARs does not relate to VCS requirements for registration.

CARs and CLs are to be resolved or closed out if the PP modifies the project description, rectifies the PD or provides adequate additional explanations or evidence that satisfies the concerns. If this is not completed, the project activity cannot be recommended for registration under VCS registry. A total of 16 findings were raised, where 05 CARs, 11 CLs and 01 FARs were raised during the validation. All the findings that are raised and communicated to project participants during the validation are included in Appendix 3.

### 2.5.1 Forward Action Requests

The Group project activity is undergoing Validation. During the validation process, an exemption has been taken by the project proponent to conduct the remote site visit of the project activity due to the significant security risk in the vicinity of project area, which has been approved by the VERRA on conditional basis/33/. A FAR has been raised by the VVB to conduct the on-site visit of the project activity during the first verification period. The details of the same can be found in Appendix 3 of this report.

## 3 VALIDATION FINDINGS

### 3.1 Project Details

The characteristics of project activity are validated are described below:

Project type, technologies and measures implemented, and eligibility of the project

The project activity “Improved Cookstove Grouped Project in Papua New Guinea” is a Grouped project with the aim of distributing fuel-efficient ICS to replace the baseline cookstoves in households in Papua New Guinea. ICS are more efficient than traditional stoves as they reduce heat loss and improve heat transfer and combustion efficiency. The project activity is distributing G3 Burn rocket stove, also known as Kuniokoa or Ecoa Wood Cookstove.

The technical parameters of G3 Burn rocket stove /9/ are:

|                    |                                                                                                                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thermal Efficiency | 50.8% (with pot skirt)                                                                                                                                                                                                                                                                               |
| Dimensions         | 32.1 cm height, 28.2 cm diameter                                                                                                                                                                                                                                                                     |
| Materials          | <ul style="list-style-type: none"> <li>▪ Stove Body: Galvalume</li> <li>▪ Pot Rest: Stainless Steel</li> <li>▪ Pot Skirt: Stainless Steel</li> <li>▪ Burning Chamber: Stainless Steel</li> <li>▪ Fuel Feeding Door: Stainless Steel</li> <li>▪ Stick Shelf: CRCA</li> <li>▪ Legs: Aluzinc</li> </ul> |
| Lifespan           | 10 Years                                                                                                                                                                                                                                                                                             |

The technology distributed under the project is G3 Burn rocket stove, having an average lifespan of 10 years and an efficiency of 50.8%, as per the water boiling test conducted by CREEC/14/. The efficiency of ICS has been proven to be more than that of traditional cookstoves. The assessment team reviewed the efficiency test report/14/ and concluded that the efficiency test was carried out in accordance with the procedure specified in the applied methodology VMR0006 v1.2/4/.

The project involves the distribution of energy-efficiency cookstoves, which falls under the category of efficiency improvements in thermal applications, thus it is eligible under the scope of the VCS Program. The project’s start date is 01/01/2024, which is based on the planned date of distribution of first ICS under this grouped project activity/02/. The crediting period is fixed for ten years, beginning on 01/01/2024 to 31/12/2033.

An agreement with the PP will be signed by the individual households using ICS confirming the use of cookstoves as well as transfer the VCU right to the PP. The sample agreement of the cookstoves was validated by the assessment team and found to be in accordance with the VCS PD/2/. The user's information collected through the End user Agreement/30/ and complied in distribution records will be used to create a record keeping system and unique identification of the ICS. This will include the user's username, identification number, address/location of residence, stove's unique serial code, and distribution date. To avoid double counting, the record keeping system will ensure that each ICS can be traced back to the project. The average estimated amount of annual emission reduction is 242,804 tCO<sub>2</sub>e and a total of 2,428,045 tCO<sub>2</sub>e for the 10-year fixed crediting period. It falls under the 'project' category as the annual average GHG emission reductions are less than 300,000 tCO<sub>2</sub>e.

The Validation team has reviewed all the above details and confirms that all the information provided is correct. CL#01, CL#02, CL#03, CL#06, CAR#01, CAR#02 and CAR#03 have been raised and resolved (Refer to Appendix 3 for further details).

#### Eligibility criteria for grouped projects

Section 1.4 of the PD/2/ clearly states the eligibility criteria, and then justifies the inclusion of project activity instances for the methodology VMR0006.

| No. | Criterion                                                                           | Compliance                                                                              | VVB Assessment                                                                                                                                                                                                                                                                                                                                 |
|-----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Meet the applicability conditions set out in the methodology applied to the project | New PIs will meet the applicability conditions set out in Section 4 of the methodology. | The validation team has reviewed the PD and the project activity is found to be fulfilling all the applicability conditions set out in section 4 of VMR0006 v1.2. Please refer section 3.3.2 of the report. New project instances(PIs) added will be checked against the applicability condition of the methodology, the condition is found to |



|   |                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                        |                                                                                                                                                                                                       | be appropriate as per section 3.6.16 of the VCS Standard/06/.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2 | Use the technologies or measures specified in the project description. | Only ICS that conforms with the grouped project description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance, | <p>The validation team determines that PP will use G3 Burn rocket stoves under this grouped project after reviewing stoves distribution database/23/, manufacturer's specifications/9/ and WBT test reports/14/, along with the remote audit/13/ interviews. The efficiency of the stove is found to be 50.8% which is greater than the baseline stove, which has been verified through the WBT test reports/14/. Any new stoves that will be added will be reviewed to satisfy the project activity applicability conditions.</p> <p>Hence, the requirements for the new project activity instances under this group project have been satisfied.</p> |

|   |                                                                                                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Apply the technologies or measures in the same manner as specified in the project description. | ICS distributed in the PIs will adhere to the grouped project description | <p>The validation team determines that G3 Burn rocket stoves under this grouped project after reviewing sales database/23/, manufacturer's specifications/9/ and WBT test reports/14/, along with the remote audit/13/ interviews, which replaces traditional cookstoves verified through the remote audit interviews/13/. The efficiency of the stove is found to be 50.8% which is greater than the baseline stove, which has been verified through the WBT test reports/14/. Any new stoves that will be added will be reviewed to satisfy the project activity applicability conditions.</p> <p>Hence, the requirements for the new project activity instances under this group project have been satisfied.</p> |
|---|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|   |                                                                                                                                           |                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.</p> | <p>New PIs will be installed only in regions within the geographic borders of PNG subject to the same baseline scenario determined in Section 3.4 in this Project Description.</p> | <p>The group project activity includes distribution of project stoves (G3 Burn rocket), which will be distributed within the boundary of Papua New Guinea.</p> <p>It was assessed during the remote audit/13/ and through the pilot distribution database/23/ that installed project stoves in the pilot phase lie within the boundary of Papua New Guinea and have the same baseline scenario i.e., the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cook stoves, as described in the VCS PD/02/. The same will be followed and checked for all new PIs that will be added for the grouped project activity.</p> |
|---|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Thus, the criteria are found to be appropriate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5 | Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances | <p>All new project activity instances will use the activity method for demonstration of additionality.</p> <p><b>Step 1: Regulatory Surplus</b></p> <p>There is no mandated government programme or policy in the host country of this project ensuring the distribution of new project activity instances.</p> <p><b>Step 2: Positive List</b></p> <p>The inclusion of new project activity instances will comply with positive list as it satisfies criterion (1) of the positive list of applied methodology where it meets all the applicability conditions of the methodology.</p> | <p>The applied methodology uses activity method for the demonstration of additionality, where it can be approved as additional considering positive list of technologies provided the stoves are distributed free of cost.</p> <p>As verified above, the geographical area of the added PAIs is the country of Papua New Guinea. There are no laws and regulations, related to cookstove distribution, applicable in the host country Papua New Guinea. The assessment team has searched the publicly available information and it is verified that there are no laws pertaining the mandate of distribution of cookstoves in Papua New</p> |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>Guinea and a letter of confirmation from PNG Forest authority has been provided to PP, which has been checked /40/.</p> <p>It was confirmed that the distribution of the household cook stoves in the Papua New Guinea is a surplus activity. There are no regulations of guidelines which mandates the project proponent to distribute the ICS.</p> <p>Further during the interviews, it was checked and confirmed with each stove beneficiary that the ICS are given to them free of cost and they are not being charged anything at the time of installation or ongoing operation.</p> <p>All the PAI installed till now comply with the additionality requirement as defined in the</p> |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                       | <p>applied methodology/04/ and in PD/02/.</p> <p>Hence, the eligibility condition is applicable.</p>                                                                                                                                                                                                                                           |
| 6  | <p>Occur within one of the designated geographic areas specified in the project description</p>                                                                                                                                                                 | <p>New PIs will only occur within the boundary of PNG.</p>                                                                                                                                                                                                                            | <p>The group project activity includes distribution of project stoves (G3 Burn rocket), which will be distributed within the boundary of Papua New Guinea. The same will be checked and for all new PIs added in the project activity.</p> <p>Hence, the eligibility condition is applicable.</p>                                              |
| 7  | <p>Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.</p>                                                        | <p>New PIs will comply with at least one complete set of eligibility criteria for the inclusion into the GPA.</p>                                                                                                                                                                     | <p>The condition is found to be appropriate to check at the time of inclusion of new PIs that they comply with at least one complete set of eligibility criteria for the GPA.</p>                                                                                                                                                              |
| 8- | <p>Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.</p> | <p>New PIs will demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body through the inclusion of sufficient technical, financial, geographic, and other relevant information within the monitoring report.</p> | <p>The PD considers the one single ICS as a Group project activity instance. The design efficiency of the project activity cook stove is confirmed from the manufacture's specification/9/.</p> <p>In view of above the expected annual energy saving for each PAI comes to approximately 0.02 GWhth/y only, which is 0.011% of the limit.</p> |

|    |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|    |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                   | <p>The efficiency of the stove is not expected to increase (or rather it will decrease) and wood (energy) saving per stove to be claimed is also fixed. So, it is concluded that no PAI has crossed or is expected to cross the limit. As none of the Group project activity instance is crossing the limit of 1% energy capacity, there is no requirement of cluster wise determination.</p>                                                    |
| 9  | Be validated at the time of verification against the applicable set of eligibility criteria.                                                                                                                                                                                        | New PIs will be validated at the time of verification against the applicable set of eligibility criteria.                                                                                         | The applicability condition is found to be appropriate to check that all the new PIs added will meet the eligibility criteria set out in PD.                                                                                                                                                                                                                                                                                                     |
| 10 | Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions) | New PIs will provide evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance. | The sample end-user agreement/30/ has been shared by the PP and it is confirmed that the VER ownership will be transferred to the PP of the project activity, same has been confirmed through the interviews conducted during the remote audit. Also, PP has shared the signed end-user agreement /47/ for the cookstoves distributed during the pilot phase to confirm the same. Hence, the applicability condition is found to be appropriate. |
| 11 | Have a start date that is the same as or later than the grouped project start date.                                                                                                                                                                                                 | The start date of new PIs will be after 01/01/2024, after the start date of the GPA.                                                                                                              | The group project activity involves distribution of G3 Burn rocket stoves in Papua New Guinea.                                                                                                                                                                                                                                                                                                                                                   |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>The start of the proposed project activity is 01/01/2024. The same has been taken as the start date of the crediting period. The crediting period for the project activity will be from 01/01/2024 to 31/12/2033 (ten years, fixed)</p> <p>The eligibility is found to be appropriate.</p>                                                                                                                                      |
| 12 | <p>Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4) and new instances are eligible for crediting from the start of the next verification period</p> | <p>New PIs will be eligible for crediting from the start date of the instance through to the end of the project crediting period (only).</p> <p>For example, when a PI starts with a fixed crediting period of 10 years (fixed), the project device/stove must be proven at inclusion that it will be able to sustain through the crediting period. This may be proven through the Project Device Lifespan Specification document.</p> | <p>The group project activity involves distribution of G3 Burn rocket stoves in Papua New Guinea. The start of the proposed project activity is 01/01/2024. The same has been taken as the start date of the crediting period. The crediting period for the project activity will be from 01/01/2024 to 31/12/2033 (ten years, fixed). The eligibility is found to be appropriate.</p>                                             |
| 13 | <p>Only eligible for crediting from the start of the verification period in which they were added to the grouped project.</p>                                                                                                                                                                                                                                                                                                                                       | <p>Crediting Periods of New PIs will only start after the date that the project has been included into the grouped project.</p>                                                                                                                                                                                                                                                                                                        | <p>The group project activity involves distribution of G3 Burn rocket stoves in Papua New Guinea. The proposed start of the project activity is 01/01/2024. The same has been taken as the start date of the crediting period. The crediting period for the project activity will be from 01/01/2024 to 31/12/2033 (ten years, fixed). New PIs that will be added on or after the start date only will be considered, and only</p> |



|    |                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                             | for the remaining duration of the crediting period. The eligibility is found to be appropriate.                                                                                                                              |
| 14 | Not be or have been enrolled in another VCS project.                                                                                           | New Pls must declare that it has not or have been enrolled in another VCS project.                                                                                                                                                                                                                                                                                                                                                          | New Pls will be checked for double counting through the declaration provided by the PP and through the secondary internet research. The eligibility condition is found to be appropriate.                                    |
| 15 | Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9 of the VCS Standard. | New Pls will adhere to the clustering and capacity limit requirements set out in Section 3.6.8 - 3.6.9 whereby the Project participant (PP) shall include all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent and where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. | New Pls under the grouped project activity will be thoroughly checked for adhering to the said requirements for multiple project activity instances set out in the VCS Standard. The eligibility is found to be appropriate. |

- Project proponent, other entities involved in the project and ownership of the VERs

The Project Proponent of this project activity is Tasman Environmental Markets Asia-Pacific Pte Ltd. As per the plan, an end user agreement/30/ will be signed by each participating household confirming that the ownership right of the carbon credits generated from this project will belong solely to the Project proponent. Earthood has checked the sample End User Agreement/30/ and verified the same. CL#07 has been raised and resolved (Refer to Appendix 3 for further details).

- Project start date

As per the VCS PD provided, the project start date is 01/01/2024 which is based on the estimated date of distribution of first ICS under this grouped project activity. The validation team certified that it complies with section 3.7 of the VCS standard, Version 4.5/6/ and is thus accepted. CAR#01 has been raised and resolved (Refer to Appendix 3 for further details).

- Project crediting period

According to VCS Standard version 4.5/6/ section 3.9.1, "The project crediting period shall be either seven years (twice renewable for a total of up to 21 years) or ten years fixed". The crediting period selected by the PP is of 10 years for the project activity from 01/01/2024 to 31/12/2033 including both days which is of fixed type and acceptable to the validation team.

- Project scale and estimated GHG emission reductions or removals

The Project is expected to reduce emissions around 242,804 tonnes of CO<sub>2</sub>e per year. The validation team confirms that the project activity falls under category 1 'Projects' according to paragraph 3.10.1 of the VCS Standard version 4.5/6/, it has the potential to reduce GHG emissions by less than 300,000 tCO<sub>2</sub>e/ year.

The average estimated amount of annual emission reduction is 242,804 tCO<sub>2</sub>e and the total of 2,428,045 tCO<sub>2</sub>e for the entire fixed crediting period of 10 years. If the underlying assumptions do not change, the emission reduction projection is reasonable, according to the validation team. CAR#02 has been raised and resolved (Refer to Appendix 3 for further details).

- Project location

The location of this project activity is the geographical boundary of Papua New Guinea (PNG), the same has been verified through the google earth application /11/ and the kml files are found to be appropriate. The KML file /11/ provided by the PP has been reviewed and it was found by the assessment team that it contains the planned sites (households) for the project activity. The file thus shared also contains the delineation of the locally administrated unit of land by the local government in PNG.

Moreover, the household chosen for pilot study/23/ is found to be within the geographical boundary of PNG. Location of New PIs will be recorded in the database and checked by the VVB during verification.

The four provinces /11/ involved as the first project instances of the group activity in Papua New Guinea.

| S. No. | Location                    | Latitude  | Longitude   |
|--------|-----------------------------|-----------|-------------|
| 1.     | Southern Highlands Province | 6.4179° S | 143.5636° E |
| 2.     | Eastern Highlands Province  | 6.5862° S | 145.6690° E |
| 3.     | Western Highlands Province  | 5.8310° S | 144.1720° E |
| 4.     | Jiwaka Province             | 5.8564° S | 144.5218° E |

- Conditions prior to project initiation

The target population's continued use of non-renewable wood fuel (firewood) to meet equivalent thermal energy needs as provided by project cookstoves in the absence of project activity is one of the prerequisites prior to project commencement.

- Project compliance with applicable laws, statutes and other regulatory frameworks

The validation team has verified that there are no laws or regulations in Papua New Guinea that restrict the use of improved cookstoves (ICS) in households, which has been checked through the letter of confirmation of PNG Forest Authority/40/. The project is a voluntary effort by the project proponent.

- Participation under other GHG programs:

Projects registered (or seeking registration) under other GHG program(s)

The project has not been registered, and it has no plans to register under any other GHG program. The VVB has checked various registries like VCS, GCC, CDM and it has been confirmed that the project is not registered any other GHG mechanism except VCS. PP has also provided the declaration/12.1/ regarding the same and is found to be appropriate.

Rejection by other GHG programs

The program has not been rejected by any other GHG program. PP has provided the declaration/12.1/ regarding the same, which states that the project has not been rejected by other GHG programs and is found to be appropriate.

- Sustainable development contributions

The project contributes to long-term development in several ways:

a. Environmental Sustainability

- a. Reduce GHG emissions
- b. Reduce the use of non-renewable biomass from forests to help conserve existing forest stock and project natural forest eco-systems and wildlife habitats

b. Social Sustainability

- a. Time saved on wood harvesting thereby reducing the workload on families and as well as presenting alternative economic development opportunities.
- b. Health benefits from reduced exposure to air pollution caused by biomass combustion
- c. Reduce accidental burn risk

c. Economic Sustainability

- a. Employment of local people throughout the project's manufacturing, distribution and monitoring.
- b. The use of the ICS will minimise household expenses on cooking

- Additional information relevant to the project, including:

Leakage management for AFOLU projects

As per the project adopts a net gross adjustment factor of 95% to account for leakage, this is not applicable.

Commercially sensitive information

No Commercially sensitive information has been excluded from the public version of the project description.

The validation team concludes that the description in the project description is accurate, complete and provides an understanding of the nature of the project. CL#04, CL#05 and CL#09 have been raised and resolved (Refer to Appendix 3 for further details).

## 3.2 Safeguards

### 3.2.1 No Net Harm

The distribution of improved cookstoves is a part of the group project activity. The project has no known or anticipated negative environmental or socio-economic consequences.

### 3.2.2 Local Stakeholder Consultation

The local stakeholder Consultation was conducted from 13/03/2023 to 20/03/2023 at 4 different venues. The participants were invited and informed about the meeting through newspaper advertisements, radio broadcasts, Letters of Notification to provincial governments, invitation letters to churches, social media posting, webinar, and email to PNG Australia Alumni Association (PNGAAA) members/17/.

| Province                    | Date       | Means of verification                                               |
|-----------------------------|------------|---------------------------------------------------------------------|
| Southern Highlands Province | 13/03/2023 | LSC Report,<br>Photographs,<br>interviews with the<br>LSC Personnel |
| Western Highlands Province  | 15/03/2023 |                                                                     |
| Jiwaka Province             | 17/03/2023 |                                                                     |
| Eastern Highlands Province  | 20/03/2023 |                                                                     |

The date and mode of invitation were confirmed from interviews of the LSC participants conducted by validation team. The date of LSC was provided in the invitation letters and advertisements. The following table provides the dates of conduction of various communication regarding the LSC:

| Communication          | Dates                 |
|------------------------|-----------------------|
| Invitation to Churches | 10/02/2023-20/03/2023 |
| Newspaper publishing   | 14 February 2023      |

|                                              |                               |
|----------------------------------------------|-------------------------------|
| <b>EHP- NOL<br/>(Letter of No-objection)</b> | 16/02/2023                    |
| <b>SHP- NOL</b>                              | 08/03/2023                    |
| <b>WHP- NOL</b>                              | 13/02/2023                    |
| <b>Jiwaka- NOL</b>                           | 15/02/2023                    |
| <b>Radio Broadcasting</b>                    | 29/02/2023<br>-<br>12/03/2023 |

Various people from different organizations/community joined the LSC and a discussion was conducted between the representative of PP and stakeholders. The participation was checked through the photographs of the LSC/18/ and attendance sheets/18/.

During interviews, it was confirmed that there were no negative comments raised during the consultation process, and participants were aware about the benefits of using Improved Cookstove over the traditional stove and users were also aware about the benefits of the improved cookstove, including lesser smoke, lesser time in wood collection, lesser wood consumption, etc. This was also confirmed through the feedback forms filled/16/ by the LSC participants. During the remote audit conducted by the assessment team, the VVB has confirmed that there was no language barrier for the local stakeholders as the project manager communicated in tok pidgin, the local language used in PNG. The same has been verified from the LSC report /19/ shared by the PP. Locals were invited through newspaper and radio advertisements /17/, to compensate for the poor internet connectivity in a few regions.

All questions raised by stakeholders were answered and documented by the PP and their representative. Stakeholder had no comments/complaints/grievances by the end of the meeting which could have any significant modification in the project description or its design. This information was cross-checked and confirmed from sample stakeholders who were interviewed by the validation team during the audit. VVB validation team was able to confirm that these stakeholders were part of the LSC meeting conducted by project developer and were found to be aware about the proceedings of the meeting. As per the VVB's assessment, PP has addressed all questions and responses of all the stakeholders with satisfaction and for future grievances PP has provided the communication details that can be used for placing the complaints after the implementation of the project.

A continuous grievance mechanism has been established and local stakeholders have confirmed that It was described during the meeting as well as the contact information of project representatives was shared with them, thus establishing a continuous grievance mechanism in conformance to the VCS guidelines/24/29/.

### 3.2.3 Environmental Impact

No Negative environmental impacts, as the project involves dissemination of Improved cookstoves, hence, environmental impact assessment (EIA) is not required for the project.

### 3.2.4 Public Comments

The project was open for public comments from 14/07/2023 to 13/08/2023. During this time, no public comments were received. CAR#02 has been raised and resolved (Refer to Appendix 3 for further details).

### 3.2.5 AFOLU-Specific Safeguards

Since, this is a non-AFOLU project activity, this section is not applicable.

## 3.3 Application of Methodology

### 3.3.1 Title and Reference

Following approved baseline & monitoring methodology is applied:

VMR0006: Methodology for installation of High Efficiency Firewood Cookstoves v1.2/04/

CDM Methodology: AMS-II.G. “Energy efficiency measures in thermal applications of non-renewable biomass” version 13.0/4/.Sectoral Scope: - 3: Energy Demand

Applied Methodological Tools:

Standard – Sampling and Surveys for CDM project activities and programmes of activities, version 09.0/21/

### 3.3.2 Applicability

The latest version of the methodology (VMR0006 v1.2 /4/) meets all the applicability conditions. As a result, the methodology is considered fully applicable for the crediting period, and no request for deviation in terms of methodology applicability is required. The VCS PD/2/ includes all the applicable conditions entirely and appropriately. The following table shows the results of the applicability conditions assessment:

| VMR0006 “Energy Efficiency and Fuel Switch Measures In Thermal Applications” v1.2                                   |               |                                                                                                            |                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methodology                                                                                                         | Applicability | Justification of Project Proponent                                                                         | Means of Validation                                                                                   |
| This methodology applies to project activities that introduce energy efficiency and fuel switch measures in thermal |               | <b>Applicable</b><br><br>The project activity involves the distribution of Improved Cookstoves whereby the | VVB conforms that the project activity is the Replacements of traditional firewood stoves with higher |

|                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>applications (including cookstoves, ovens, and dryers) that:</p> <ol style="list-style-type: none"> <li>1. Increase thermal efficiency to reduce the consumption of non-renewable biomass; or</li> <li>2. Switch from fossil fuel (coal or kerosene) to renewable biomass in new or existing improved thermal energy generation units.</li> </ol>                                                                        | <p>cookstoves have an increased thermal efficiency to reduce the consumption of non-renewable biomass, therefore this criterion is met.</p>                                                                                                                                  | <p>efficiency stoves in the targeted households as confirmed by remote-site interviews of the PP and the end-users/13/. The grouped project activity is the distribution of G3 Rocket stoves. The efficiency of the stove is found to be 50.8% which is greater than the baseline stove, which has been verified through the WBT test reports/14/. Hence, the applicability condition is found to be met.</p> |
| <p>This methodology is applicable to both 'Projects' and 'Large Projects' under the following conditions:</p> <ol style="list-style-type: none"> <li>1. All applicability conditions of the latest version of AMS II.G. must be met.</li> <li>2. The project activities must be implemented in households, community-based kitchens, institutions (e.g., schools), or small and medium-sized enterprises (SMEs).</li> </ol> | <p><b>Applicable</b></p> <p>The project activity is a project that will meet all applicability conditions of the latest version of AMS II.G. v13.0/04/ (see table below), and be implemented in households within the project boundary, therefore this criterion is met.</p> | <p>VVB confirms that grouped project activity meets all the applicability conditions of the latest version of AMS II.G as demonstrated in the table below and will be distributed in households within the boundary of PNG, through remote interview. Hence, applicability condition is found to be met.</p>                                                                                                  |
| <p>For fuel switch activities, the following additional conditions must be met:</p> <ol style="list-style-type: none"> <li>1. Projects must exclusively use renewable biomass, and meet the following additional conditions:</li> </ol>                                                                                                                                                                                     | <p><b>Not Applicable</b></p> <p>The project activity involves the distribution of improved cookstoves with increased thermal efficiency to reduce the consumption of non-renewable biomass and is not considered a fuel switch activity according to the</p>                 | <p>The project activity is the Replacements of traditional firewood stoves with higher efficiency stoves in the targeted households and doesn't involve the fuel switch measures. Hence, the condition is not applicable.</p>                                                                                                                                                                                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                 |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--|
| <p>a. If biomass residues are used, they have been left for decay or burned without energy recovery before the implementation of the project activity.</p> <p>b. If biomass residues from a production process are used, the implementation of the project does not result in an increase of the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.</p> <p>c. If biomass from dedicated plantations is used, the applicability conditions of TOOL16 “Project and leakage emissions from biomass” must be satisfied.</p> | <p>methodology, therefore this criterion is not applicable.</p> |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--|



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| <p>2. The renewable biomass sources must be documented in the project description and monitoring periods, including origin, quantities, and pre-project conditions. If the biomass is sourced from a third-party, proof of purchase must be provided (e.g., contractual agreements or purchase receipts).</p> <p>More than one type of biomass may be used (e.g., briquettes and wood chips).</p> |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                  |
| <p>VMR0006 must be used in conjunction with the latest version of AMS-II.G. All the procedures and requirements of AMS-II.G. must be applied unless VMR0006 indicates otherwise.</p>                                                                                                                                                                                                              | <p><b>Applicable</b></p> <p>The project activity will adhere to the latest versions of VMR0006 and AMS-II.G, whereby all requirements of AMS-II.G. will be applied unless VMR0006 indicates otherwise.</p> | <p>The validation team has checked the requirements VCS PD and confirms that project activity adhere to the procedures and requirements set out in VMR0006 v1.2 and AMS-II.G v13.0. Hence, the applicability is found to be fulfilled.</p>                                                                                       |
| <p>CDM Methodology: AMS-II.G. v13.0 “Energy efficiency measures in thermal applications of non-renewable biomass”</p>                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                  |
| <p>In the case of cookstoves, the methodology is applicable to the introduction of single pot or multi pot portable or in-situ cookstoves with rated efficiency of at least 25 per cent.</p>                                                                                                                                                                                                      | <p><b>Applicable</b></p> <p>The thermal efficiency of the stoves is 50.8%, which exceeds the required thermal efficiency of at least 25%.</p>                                                              | <p>VVB confirms that the G3 Burn rocket stove has been tested for thermal efficiency of 50.8% as per water boiling test (WBT 4.2.3) certified by CREEC/14/. Therefore, in accordance with methodological standards, the project stoves have a stipulated high power thermal efficiency of at least 25% as confirmed from the</p> |

|                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | manufacturer's specifications/9/. Hence, the applicability condition is found to be met.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>The aggregate energy savings of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input.</p> | <p><b>Applicable</b></p> <p>The maximum number of improved cookstoves is defined according to the corresponding stove performance, to ensure a maximum energy saving of 180GWh/year. Any additional emission savings will be counted within a new project instance.</p> <p>The first project instance covers the 4 highlands provinces in PNG which reportedly has 535,354 households<sup>1</sup>.</p> <p>According to ER Calculations, the maximum number of cookstoves deployable before reaching the 60GWh/year limit per PAI is 1,214,064 Cookstoves, therefore the first PAI meets the applicability criteria.</p> <p>In the event a Project Instance exceeds the 60GWh/year limit, a new Project Instance must be included.</p> | <p>The validation team has confirmed that the energy savings of single project activity instances is not greater than 60 GWh per year or 180 GWh thermal per year, through the estimated calculations of energy saving provided in the ER sheet, the calculation has been reviewed and the formulas used are found to be correct. The PP has properly defined the project instance which will be the case where the limit of 60 GWh/ year will not be crossed by the one project instance. In the event the 60Wh/year limit is crossed, a new PAI will be included. This has been found to be appropriate.</p> |
| <p>Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods</p>                                                  | <p><b>Applicable</b></p> <p>Using published literature, it can be proven that non- renewable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Based on the review of evidence provided by the PP /35//36//37/, VVB confirms that non-</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

<sup>1</sup> <https://www.nso.gov.pg/census-surveys/national-population-housing-census/>

|                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| or referring to published literature, official reports or statistics.                                                                                                                                                                                               | <p>biomass has been used in the project region since 31 December 1989, therefore this criterion is met.</p> <p>Refer to the end of this section for full explanation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | renewable biomass was being used in the project region since 31 December 1989.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| For cases where the biomass is sourced from renewable sources, the project participants should use a corresponding Type I methodology.                                                                                                                              | <p><b>Not Applicable</b></p> <p>The Project Activity is not a fuel switch project; therefore this criterion is not applicable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | The grouped project activity is the distribution of improved cookstove which will utilize firewood as a source of fuel, hence, condition is not applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo). | <p><b>Applicable</b></p> <p>In order to avoid double counting of emission reductions through the method of distribution, each ICS will carry a Unique Identifier Number (UIN) and the distribution of each device is logged into an electronic database which includes unique information for each recipient (i.e., Unique Identification Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no UINs are double counted.</p> <p>During the monitoring period, survey samples will be only taken from the UINs that are in the electronic database of end-users eliminating the risk of double counting from any other projects within the same geographical location.</p> | The validation team has confirmed through audit interviews/13/ that the suggested method for project device distribution includes measures to prevent double counting of emission reductions, such as unique product identification, end user information (name, address, etc.), and unique GPS referenced location (if available). The validation team has thus confirmed that the record-keeping system will prevent double counting. The VVB has also cross-checked various registries such as UNFCCC CDM, Gold Standard and Verra for similar kind of projects and it has been confirmed that there are no similar projects identified in the project boundary. |

|                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.</p> | <p><b>Applicable</b></p> <p>A clause in the commercial contract will be in place for the start of the project (01/012/2023) that clearly states the manufacturer will not have any claim over the carbon rights from the cookstoves, hence eliminating the possibility of double counting.</p> | <p>The validation team has confirmed during the remote audit interviews with the PP /13/ that the proposed methods will be applied which includes the signing of the commercial contract having the clause about surrendering any claim over the carbon rights from the cookstoves. It will prevent double counting of emission reductions VVB has also reviewed the invoice for 160 cookstoves (for pilot project), which sells the product in its entirety with all rights attached to the product, which confirms that the emission reductions will only be claimed by the project activity.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Evidence that the non-renewable biomass has been in use since 1989

As demonstrated by a 2013 report from ACIAR /37/ that 8.80 million people of PNG consumes well over 9.40 million tonnes of firewood per year, also as per the IRENA report of 2013 /46/, over 90.00% of the population uses firewood for domestic and commercial cooking on open fires in Papua New Guinea hence, fuelwood has been in use since 31 December 1989 for major domestic energy consumption. It further shows that fuelwood is popular as the primary energy source for cooking and heating especially in the highlands region of Papua New Guinea (PNG). It is noteworthy that 40% of the island's population lives in this region, making it a vital part of the domestic energy market. A baseline survey was also conducted by a third-party consultant team in October - November 2022 /31/ (Cookstoves Highlands region Baseline report v2.pdf). It was evident that in the baseline, the major population in Papua New Guinea use firewood along with the use of traditional cookstoves, which is further supported by several studies in the report (like ACIAR, 2013/37/, IRENA report of 2013/46/ and Page et al. (2016)/41/).

The assessment team has assessed the applicability of the said methodology and concluded that it is correct. CL#08 and CAR#05 were raised and resolved (Refer to Appendix 3 for further details).

### 3.3.3 Project Boundary

According to VMR0006 v1.2, "The project boundary must be determined following the procedure provided in the latest version of AMS-II.G.". PP has determined the project boundary in line with the AMS-II.G as follows:

According to AMS-II.G. “Energy efficiency measures in thermal applications of non-renewable biomass” Version 13.0/4/, the Project Boundary is established. The greenhouse gas sources identified in the VCS PD/2/ are deemed appropriate and evaluated below:

The project boundary includes the physical and geographical locations of the distributed ICS (G3 Burn rocket stove) inside boundary of Papua New Guinea for the aggregated Project.

The validation team was able to ensure that the approved methodology/4/ addresses all the identified emission sources that are impacted by the project activity, as shown in Table below.

| Source   |                                                        | Gas              | Included? | Justification/Explanation  |
|----------|--------------------------------------------------------|------------------|-----------|----------------------------|
| Baseline | Emission from use of non-renewable biomass/Fossil fuel | CO <sub>2</sub>  | Yes       | Major source               |
|          |                                                        | CH <sub>4</sub>  | Yes       | Major source               |
|          |                                                        | N <sub>2</sub> O | Yes       | Major source               |
|          |                                                        | Other            | No        | No other source identified |
| Project  | Emission from use of non-renewable biomass             | CO <sub>2</sub>  | Yes       | Major source               |
|          |                                                        | CH <sub>4</sub>  | Yes       | Major source               |
|          |                                                        | N <sub>2</sub> O | Yes       | Major source               |
|          |                                                        | Other            | No        | No other source identified |

The greenhouse gas sources indicated in the PD/2/ are transparently stated and deemed appropriate and consistent with the technique used. The validation team verifies that the project boundary, as well as the selected sources, sinks, and reservoirs, are all justified. CAR#01 was raised and resolved (Refer appendix 3 for further details).

### 3.3.4 Baseline Scenario

PP has conducted the baseline survey to justify the use of traditional cookstoves by the households in the absence of the project activity and it is observed that there was predominance in the use of traditional cookstoves.

The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of

project activity. The same has been verified while interviewing the end users during the remote audit/13/.

A baseline survey has been conducted by a third-party consultant; report of baseline survey has been shared by the PP. The study conducted has been further supported by the published literature (like ACIAR, 2013, IRENA report of 2013 and Page et al. (2016))/37/41/46/, which has been checked and the baseline study conducted is found to be appropriate, which suggests that the consumption of up to 10 - 13 million tonnes of firewood per year, where predominant source of fuel for cooking purposes is the firewood with open fire cookstove. The survey also showed that 81.00% and 90.00% of the households in the Western Highland Province and southern Highlands Province use traditional Open fire cookstove with firewood.

The procedures for identifying the baseline scenario have been correctly followed and the identified scenario reasonably represents what would have occurred in the absence of the project. VVB confirms that the assumption and data can be deemed reasonable. CL#04, CL#09 and CL#11 were raised and resolved (Refer appendix 3 for further details).

### 3.3.5 Additionality

The additionality is defined based on the methodology VMR0006 v1.1/4/ as below:

#### Activity Method

##### Step 1: Regulatory Surplus

The VVB investigated and confirmed that no governmental programme or policy in the host country mandates the distribution of household fuel-efficient cookstoves/40/. As a result, the project is not mandated by any law, legislation, or regulatory framework, or by any systematically implemented law, statute, or other regulatory structure for UNFCCC non-Annex I nations.

Thus, the participation is purely voluntarily. The VVB confirms that the proposed project is a voluntary coordinated action by Tasman Environmental Markets Asia-Pacific Pte Ltd.

##### Step 2: Positive List

In accordance with section 3.3.2 of this report, the project fits all the applicability requirements of the methodology/4/ which represents the positive list. The PP has demonstrated (with the help of two evidence, viz, TEM's end user agreement for zero cost /48a/, and TEM funding letter for exclusion from government schemes and multilateral funds /48b/) that the current project activity comply with the positive list conditions 1 & 2. The VVB has reviewed the documents mentioned above and found them reasonable and acceptable. Hence, it has been confirmed that the applicability conditions for positive list stated in section 7 of VMR0006/04/ has been met.

It was confirmed through the interviews/13/ of the selected users from database of the pilot project implementation and the local stakeholders, if they were charged anything for the stoves. In response to the question, it was clearly mentioned by the end users and the local stakeholders that the stoves were provided to them completely free of cost.

The same was also discussed with the PP and the implementation partners during the closing meeting of audit that the ICS has been installed at no cost to the household and the only source of revenue is the selling of GHG credits. Also, the project is not part of any government initiative or backed by multilateral funds. The PP has signed the end-user agreement for zero cost /48a/ and funding letter for exclusion from government schemes and multilateral funds /48b/ to demonstrate the same.

Therefore, the validation team confirms that the proposed project activity falls into the positive list criteria as per the applied methodology VMR0006/04/.

### Step 3: Project Method

VVB determined and confirmed through the interviews/13/ and declaration of the PP/12.3/ that the ICS is installed at zero cost to the households and the project's only source of revenue is the sale of GHG credits.

As the stoves are provided at zero cost to the end-users and has no other source of revenues other than the sale of the GHG credits, the project activity has not applied the investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible.

Conclusion: The project is considered additional because it satisfies the positive list criteria of the applied methodology/4/ (VMR0006 version 1.2) for the demonstration of additionality.

### 3.3.6 Quantification of GHG Emission Reductions and Removals

The applied methodology VMR0006 does not account for baseline emissions and project emissions separately, however, quantifies net emission reductions achieved by the project.

#### Leakage:

As per section 8.3 of VMR0006 v1.2, "Projects must apply an adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity ( $Adj_{LE}$ ). The procedures from the latest version of AMS-II.G. must be applied.  $Adj_{LE}$  is fixed at validation."

As per para 41 of AMS-II.G v13.0, "Leakage related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The potential source of leakage due to the use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources shall be considered. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass by the non-project households/users, that is attributable to the project activity, then *Bold<sub>i,j</sub>* is adjusted to account for the quantified leakage. Alternatively, *By,savings<sub>i,j</sub>* is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required."

PP has chooses to apply the leakage factor 'Adj<sub>LE</sub>' by multiplying a net gross adjustment factor of 0.95. the applied approach is in line with the applied methodology VMR0006 v1.2 and AMS-II.G v13.0, and hence, found to be appropriate.

Net Emission Reductions:

As per the applied methodology/4/, net emission reductions are calculated by applying following equations, for project activities replacing baseline stoves using non-renewable biomass (firewood/ charcoal):

$$ER_y = \sum \sum B_{y,savings,i,j} \times N_{0,i,j} \times n_{y,i,j} \times NCV_{biomass} \times u_d \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times Adj_{LE} \times (1 - u_d)$$

Where,

|                     |   |                                                                                                                                                         |
|---------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| $B_{y,savings,i,j}$ | = | Quantity of woody biomass that is saved in tonnes per improved cookstove of type <i>i</i> and batch <i>j</i> during year <i>y</i>                       |
| $N_{0,i,j}$         | = | Number of project devices of type <i>i</i> and batch <i>j</i> commissioned (number)                                                                     |
| $n_{y,i,j}$         | = | Proportion of commissioned project devices of type <i>i</i> and batch <i>j</i> ( $N_{0,i,j}$ ) that remain operating in year <i>y</i> (fraction)        |
| $f_{NRB,y}$         | = | Fraction of woody biomass that can be established as non-renewable biomass (fNRB)                                                                       |
| $NCV_{biomass}$     | = | Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne) <sup>2</sup>        |
| $EF_{wf,CO_2}$      | = | CO <sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO <sub>2</sub> /TJ) <sup>3</sup>       |
| $EF_{wf,non\ CO_2}$ | = | Non-CO <sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO <sub>2</sub> /TJ) <sup>4</sup> |
| $Adj_{LE}$          | = | Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (fraction)                            |
| $u_d$               | = | Uncertainty deduction for fnrb (%)                                                                                                                      |

As per the section 8.4 of the applied methodology, VMR0006 v1.2,  $B_{y,savings,i,j}$  must be determined by applying the procedures and requirements of AMS-II.G. PP is using option 3 : Water boiling test to

<sup>2</sup> 2019 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

<sup>3</sup> 2019 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

<sup>4</sup> 2019 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion



determine the same, in line with the requirements stated in AMS-II.G and hence, found to be appropriate. The quantity of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{old,i,j} \times \left( 1 - \frac{\eta_{old,i,j}}{\eta_{new,i,j}} \right)$$

Where,

- $\eta_{old}$  = Efficiency of the old devices being replaced by project devices of type i and batch j (fraction)
- $\eta_{new,y,i,j}$  = Efficiency of the project device i and batch j (fraction)
- $B_{old,i,j}$  = Annual quantity of woody biomass that would have been used in the absence of the project activity to generate thermal energy equivalent to that provided by the project device type i and batch j (tonnes/year)

Following are the assumptions for ex-ante calculations:

1. Approximately 100,000 ICS, namely, G3 Burn rocket stove will be installed under this project.
2. The annual rate of stove loss is estimated to be 5% for G3 Burn rocket stove.
3. The value of  $B_{old,i,j}$  is fixed as 5.56 tonnes/ year as per PP's baseline study/39/ and  $f_{NRB,b,y}$  is fixed as 0.81 as per PP's literature review and baseline calculations.

Sample calculation:

$$B_{y,savings,i,j} = B_{old,i,j} \times \left( 1 - \frac{\eta_{old,i,j}}{\eta_{new,i,j}} \right)$$

$$B_{y,savings,i,j} = 5.56 \times (1 - 0.15/0.508) = 3.92 \text{ tonnes/ year}$$

The year wise estimated Emission reduction calculated are as follows:

| Year | $B_{y,savings,i,j}$ | $N_{o,i,j}$ | $\eta_{y,i,j}$ | $\mu_y$ | $f_{NRB,y}$ | $NCV_{biomass}$ | $EF_{wf,CO2} + EF_{wf,nonCO2}$ | $Adj_{LE}$ | $1 - \mu_d$ | $ER_{y,i,j}$ |
|------|---------------------|-------------|----------------|---------|-------------|-----------------|--------------------------------|------------|-------------|--------------|
| 1    | 3.81                | 40,000      | 0.95           | 0.90    | 0.81        | 0.02            | 121.46                         | 0.95       | 0.74        | 140,716.61   |
| 2    | 3.80                | 80,000      | 0.90           | 0.90    | 0.81        | 0.02            | 121.46                         | 0.95       | 0.74        | 266,124.72   |
| 3    | 3.78                | 100,000     | 0.86           | 0.90    | 0.81        | 0.02            | 121.46                         | 0.95       | 0.74        | 251,631.60   |
| 4    | 3.76                | 100,000     | 0.81           | 0.90    | 0.81        | 0.02            | 121.46                         | 0.95       | 0.74        | 297,388.88   |
| 5    | 3.74                | 100,000     | 0.77           | 0.90    | 0.81        | 0.02            | 121.46                         | 0.95       | 0.74        | 281,153.30   |

|                        |      |         |      |      |      |      |        |      |      |              |
|------------------------|------|---------|------|------|------|------|--------|------|------|--------------|
| 6                      | 3.72 | 100,000 | 0.74 | 0.90 | 0.81 | 0.02 | 121.46 | 0.95 | 0.74 | 265,784.70   |
| 7                      | 3.70 | 100,000 | 0.70 | 0.90 | 0.81 | 0.02 | 121.46 | 0.95 | 0.74 | 251,237.50   |
| 8                      | 3.69 | 100,000 | 0.66 | 0.90 | 0.81 | 0.02 | 121.46 | 0.95 | 0.74 | 237,468.48   |
| 9                      | 3.67 | 100,000 | 0.63 | 0.90 | 0.81 | 0.02 | 121.46 | 0.95 | 0.74 | 224,436.69   |
| 10                     | 3.65 | 100,000 | 0.60 | 0.90 | 0.81 | 0.02 | 121.46 | 0.95 | 0.74 | 212,103.29   |
| <b>Total:</b>          |      |         |      |      |      |      |        |      |      | 2,428,045.79 |
| <b>Annual Average:</b> |      |         |      |      |      |      |        |      |      | 242,804.58   |

The equations stated above under section 4.4 of the PD were compared with the applied methodology/4/ and were found to be exactly same. The same equations have been used in the estimated ER sheet/3/ to calculate the estimated ERs for the entire crediting period. CL#10 and CAR#04 have been raised and resolved (Refer Appendix 3 for further details).

Thus, the validation team confirms that the project activity complies with the specified requirements and formulae used to determine emission reductions as discussed above. The assessment team also confirms that:

- All assumptions and data used by the PP are listed in the VCS PD, including their references and sources;
- All documentation used by the PP as the basis for assumptions and source of data is correctly quoted and interpreted in the VCS PD;
- The baseline methodology has been applied correctly to calculate baseline emission, whereas justification for project emissions, leakage emissions, etc., are provided correctly and ex-ante emission reductions are demonstrated;
- All estimates of the ex-ate proposed baseline emissions can be replicated using the parameters and proposed values provided in the VCS PD.

### 3.3.7 Methodology Deviations

The VMR0006 v1.2 in conjunction with AMS-II.G. version 13.0/4/ was adopted for the project activity, and the PP adhered to the VCS PD development process. The validation team has confirmed that the project activity has not been subjected to any methodology deviations.

### 3.3.8 Monitoring Plan

The project's monitoring plan is in compliance with the applied methodology, VMR0006 v1.2 /4/. PP has taken the monitoring parameters from AMS-II.G v13.0 along with some additional parameters from VMR0006, which is found to be inline with the applied methodology VMR006 v1.2 and hence,

found acceptable. The monitoring plan will give an opportunity for real measurement of emission reductions achieved. The monitoring plan for the same has been included in section 5 of the VCS PD/2/. Validation team confirms that the project participant is able to implement the monitoring plan.

Data and parameters fixed ex-ante:

| Parameter           | Description                                                                 | Value | Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fNRB <sub>b,y</sub> | Fraction of woody biomass that can be established as non-renewable biomass. | 0.81  | <p>The fraction of woody biomass saved is calculated using CDM tool 30/20/. The value of the parameter has been audited by a third-party scientific expert, letter of approval has been shared by the PP and is found to be appropriate.</p> <p>(supportive files titled: GeoNarem Letter for fNRB_Jan 23.pdf (third-party forested that prepared the fNRB calculations) and SPFEC Letter to TEM-Review of FnRB Calculations.pdf (third-party auditor). The calculation sheet/25/ has also been shared and the reference data to calculate, equations and approach is found to be in line with methodological tool 30/20/. The value has been calculated using Papua New Guinea Forest Authority (2019), Forest and Land Use Change in PNG 2000 – 2015, and Climate Change and Development Authority (2022). PNG's Second Biennial Update Report to UNFCCC. Papua New Guinea 2021 National Report, FAO (2020). Global Forest Resource Assessment, (FRA) PNG Report/43/44/45/, the reference taken has been reviewed and found to be appropriate. PP has also taken reference from the old UN Default values 99% and the current value of 87.82% using UN Default Value Guidelines (EB67, Annex 22)/34/ and the selected value has been</p> |

|                     |                                                                                                                                          |                           |                                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                                                                                                                                          |                           | found to be most conservative taken. The value has been consistently reported in the VCS PD/2/. VVB has cross-checked the value of $f_{NRB}$ with the CDM project (CDM PoA 10613), where the value is 94%, hence, the value calculated by the PP is found to be more conservative and hence, acceptable. |
| $NCV_{wood, fuel}$  | Net calorific value of the non-renewable woody biomass, renewable biomass, briquettes, or pellets used in project devices                | 0.0156 TJ/tonne           | As per the applied methodology/4/, the net calorific value of non-renewable woody biomass that is substituted or reduced, is taken from the Guidelines for 2019 that are a refinement to 2006 IPCC National Greenhouse Gas Inventories; Volume 2 Energy/39/.                                             |
| $EF_{wf, CO_2}$     | CO <sub>2</sub> emission factor of biomass that is substituted or reduced                                                                | 112 tCO <sub>2</sub> /TJ  | The value taken for CO <sub>2</sub> emission factor for the use of wood is a Default value from 2019 Refinement to 2006 IPCC guidelines/39/. The value has been consistently reported in the VCS PD/2/.                                                                                                  |
| $EF_{wf, non-CO_2}$ | Non-CO <sub>2</sub> emission factor of biomass that is substituted or reduced                                                            | 9.46 tCO <sub>2</sub> /TJ | The value used for non-CO <sub>2</sub> emission factor for the use of wood fuel is taken from 2019 Refinement to 2006 IPCC Guidelines/39/. The value has been consistently reported in the VCS PD/2/.                                                                                                    |
| Life Span           | The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the | 10 Years                  | The value used has been verified from the manufacturing specifications of the improved cookstove/9/.                                                                                                                                                                                                     |

|                  | project stove<br>efficiency                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|--------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\eta_{old}$     | Efficiency of<br>baseline<br>cookstove                                                                             | 0.15  | PP has applied the default value of three-stone firewood with no improved combustion air supply. PP has conducted the baseline study during which it has been determined that the pre-project device is the traditional three-stone fire with no improved combustion air supply or flue gas ventilation, the same has been reviewed through the baseline study report/31/, and baseline study sheets/10/ submitted by the PP (titled TEM 2023 Cookstoves Highlands region Baseline survey report v2). The value for efficiency of baseline cookstove is taken from CDM Tool 33 /38/ and is consistently reported in the VCS PD/2/. The same has been verified during the interviews/13/ conducted and is found to be appropriate. |
| $\eta_{new,i,j}$ | Efficiency of<br>project stove                                                                                     | 0.508 | The value for this parameter is taken from the manufacturer's specification/09/, Thermal Efficiency Test Report/14/ and is consistently reported in the VCS PD/2/.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ADjLE            | Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity. | 0.95  | The default value has been taken from the latest version AMS-IL.G, which is found to be in line with the applied methodology/04/ and is consistent with the VCS PD/02/.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ud               | Uncertainty deduction for<br>fnrb                                                                                  | 0.26  | The default value of the parameter has been taken from applied methodology VMR0006/04/. Since, the value has been calculated as per tool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|      |                                                                                                                                                                                                          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                                                                                                          |                     | 30/20/, the value has been found in line with the applied methodology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bold | Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices. | 5.56<br>Tonnes/year | <p>PP has conducted the baseline survey (TEM (2023). Cookstoves Highlands region baseline survey report v2/31/). in WHP and SHP to determine the value of the parameter and taken the value from the survey questionnaires considering the preparation of 2 meals daily, which has been reviewed and found to be appropriate and in line with the conditions stated in VMR0006 v.1.2. The value considered has been cross-checked with the CCT conducted by the PP at a later stage to understand the requirements of the locals and the efficiency and performance of the distributed stoves. It has also been observed that the value applied by the PP is more conservative than the value obtained by through the CCT. The value has been audited by SPFEC and is submitted by the PP under the name of SPFEC Letter_Cookstoves Highlands region baseline report v2.pdf/31/. It has also been cross-checked with the peer-reviewed research studies in PNG/37/41/42/ and the value is found to be appropriate. The value has been consistently reported in the VCS PD/2/.</p> <p>The baseline survey has been conducted in October 2022, and at the time of the survey, VMR0006 v1.1 was latest available version of the methodology. In the previous version of VMR0006 (v1.1), a confidence/precision analysis</p> |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>for sample sizes wasn't a requirement. As per VMR0006 v1.1, for a population size of greater than 1000, PP can choose 100 random samples from the entire population. The PP carried out 220 random samples to maintain the accuracy of the data gathered from the surveys and minimize the impact of any anomalies within the sample.</p> <p>When project validation commenced in July 2023, a new version of the methodology (VMR0006 v1.2) was introduced, establishing a compulsory confidence/precision threshold of 90/10. PP has verified the confidence and precision of the sample population as per the requirement, by using 5.56 as the estimated expected mean of with a standard deviation of 4.00, extracted from the data presented in the baseline survey, leading to an expected number of samples of 141. PP has used the CDM Sample calculator /50/, to calculate these sample size, which is found to be appropriate and aligned with the current methodology requirements. The PP implemented a Simple Random Sampling by Mean (SRS-Mean) technique /50/ which has been found in accordance with the CDM Standard for sampling and surveys for CDM project activities and programme of activities, version 9.0 /21/ and CDM Sampling Guidelines/22/. Similarly, the precision has been calculated by the PP for the 220 samples that has been</p> |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |  |  |                                                                                                                                                          |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | conducted to be 8.00%, which is found to be under the 10% precision, and hence, the 90/10 confidence/ precision has been found to be met by the PP /50/. |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------|

Data and parameter to be monitored:

| Parameter   | Description                                                                                                     | Monitoring Frequency                         | Assessment                                                                                                                                                                                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $No_{i,j}$  | Number of project devices of type i and batch j commissioned                                                    | Yearly, when there are new commissioned ICSs | The parameter will be verified through the stove distribution records and purchase invoice of the improved cookstoves. The parameter will be used to calculate the number of devices operational during the monitoring period.                                                                                                                         |
| $n_{y,i,j}$ | Proportion of commissioned project devices of type i and batch j ( $No_{i,j}$ ) that remain operating in year y | Biennially                                   | The parameter will be monitored through the monitoring surveys conducted by the PP. This will be measured based on a representative sample to achieve 95/10 confidence precision, the approach is found to be in line with AMS-II.G and hence, found appropriate.                                                                                      |
| $\mu_y$     | Adjustment to account for any continued use of pre-project devices during the year y                            | Biennially                                   | The parameter will be used to check the usage of baseline stoves or pre-project devices in the households, where improved cookstove is distributed. The value of the parameter will be determined through the monitoring surveys conducted by the PP and the emission reductions are adjusted based on the continued usage of pre-project devices. The |



|                                           |                                                                                                                                                                                                                                             |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                                                                                                                                                                                                                             |                                                                                                       | approach defined has been reviewed and is found to be inline with AMS-II.G and hence, appropriate.                                                                                                                                                                                                                                                                                                                                                                                      |
| $\eta_{new,i,j}$                          | Efficiency of the device of each type i and batch j implemented as part of the project activity                                                                                                                                             | Yearly                                                                                                | The parameter will be used to monitor the efficiency of the project device. The value of the parameter will be determined based on the manufacturer's specifications which will be based on WBT test conducted on the cookstove, which is found to be in line with AMS II.G. PP has also applied the loss in efficiency for the parameter throughout the vintage of cookstove, which will be determined through the AMS II.G para 37 option (b) or (c). Hence, found to be appropriate. |
| Date of commissioning of Batch j          | To establish the date of commissioning, the Project Participant may opt to group the devices in "batches" and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch | Fixed and recorded at the time of commissioning/ distribution of the last project device in the batch | The parameter will be used to monitor the date of commissioning of each batch. PP will record the commissioning of each batch in the project database. The value of the parameter will be recorded as the last date of project device to be registered under each batch.                                                                                                                                                                                                                |
| Date of commissioning of Project Device i | Actual date of commissioning of the project device                                                                                                                                                                                          | Fixed and recorded at the time of commissioning/ distribution                                         | The parameter will be used to monitor the date of commissioning of each project device. PP will record the commissioning                                                                                                                                                                                                                                                                                                                                                                |

|  |  |  |                                                                                                             |
|--|--|--|-------------------------------------------------------------------------------------------------------------|
|  |  |  | of each project device in the project database and will be recorded during the registration of each device. |
|--|--|--|-------------------------------------------------------------------------------------------------------------|

All documentation and records must be retained safely and easily accessible for at least two years after the project crediting period ends, in line with section 3.25.1 of the VCS Standard (v.4.5)/6/. The validation team accepts the data collection and administration techniques described in section 5.3 of the VCS PD/2/. The validation team spoke with PP representatives, and it was determined that PP creates and maintains a database of all the project equipment it delivered.

According to the validation team, the monitoring plan's implementation tools, such as data management, monitoring tools, and quality assurance and quality control procedures, are adequate to ensure that the emission reductions achieved by or resulting from the grouped project proposal therein can be reported ex-post and verified. The sampling plan also complies with the monitoring methodology VMR0006 (v.1.2)/4/, the Standard of Sampling and Surveys of CDM Project Activities and Programme of Activities (v.09.0)/21/, and the Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities (v.04)/22/ to estimate the monitored parameter values used in the ER calculations. The target population will consist of the end-users in rural/ semi-urban areas listed in the database on which Simple Random Sampling will be applied. The sample size equations stated in the PD/1/ were checked and found consistent with CDM Guidelines: Sampling and Survey for CDM project activities and programmes of activities, version 4/22/. The sampling frames will include distributed stoves based on the total population of ICS for the parameters Proportion of commissioned project device that remains operating ( $n_{y,i,j}$ ) and continued use of pre-project devices ( $\mu_y$ ). The PP will apply a 95/10 confidence/precision for annual and 95/5 confidence/precision in case of biennial, since this is a large-scale project activity, which is found to be in line with the CDM Guidelines: Sampling and Survey for CDM project activities and programmes of activities, version 4/22/.

CL#05, CL#12 and CAR#02 was raised and resolved (Refer Appendix 3 for further details).

The validation team certifies that the overall monitoring plan complies with the methodology VMR0006 (v.1.2)/4/ requirements, that the monitoring arrangements described in the monitoring plan/2/ are practical within the project design, and that the project proponents will be able to carry out the described monitoring plan/2/.

### 3.4 Non-Permanence Risk Analysis

Since, this is a non-AFOLU project, this section is not applicable.

## 4 VALIDATION OPINION

Earthood Services Private Limited (hereafter referred to as Earthood) has been contracted by Tasman Environmental Markets Asia-Pacific Pte Ltd to conduct the VCS validation of the project activity – “Improved

Cookstove Grouped Project in Papua New Guinea”, with VCS ID 4416, in accordance with the relevant requirements of VCS Standard version 4.5/6/; to achieve real, measurable, additional, and permanent emission reductions.

The proposed project aims to improve household and community access to improved cookstoves by distributing high thermal efficiency, low greenhouse gas emission cooking stove known as improved cookstoves to households of Papua New Guinea. The replacement of inefficient traditional cook stoves through ICS will help to reduce greenhouse gas emissions and firewood use, resulting in long-term climate change mitigation.

This validation was carried out using a risk-based approach. 05 Corrective action Requests (CAR), 12 Clarification Requests (CLs) and 01 Forward Action Requests (FARs) were raised and successfully closed during the validation.

The validation is based on the additional documents pertaining to baseline and monitoring methodology/4/, as well as follow-up interviews/13/ and supporting documents made available to the validation team by the PP.

The validation team is of the opinion that the information provided in PD/2/ fulfills the criteria of VCS Standard version 4.5/6/ and is in line with all the relevant VCS requirements. The validation has been performed using a risk-based approach as described above. The project emission reductions are calculated in a transparent and conservative manner, so that the expected average estimated amount of annual emission reduction is 242,804 tCO<sub>2</sub>e and a total of 2,428,045 tCO<sub>2</sub>e for 10 years of fixed crediting period and in accordance with the applied methodology VMR0006 Version 1.2, VCS standard Version 4.5 and ISO 14064-3.

Validation period must be broken down into calendar year vintages: From 01/01/2024 to 31/12/2033.

Validated GHG emission reductions and removals in the above period:

| Year                                 | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|--------------------------------------|--------------------------------------------------------------------|
| (01/01/2024 to 31/12/2024)<br>Year 1 | <b>140,716.61</b>                                                  |
| (01/01/2025 to 31/12/2025)<br>Year 2 | <b>266,124.72</b>                                                  |
| (01/01/2026 to 31/12/2026)<br>Year 3 | <b>251,631.60</b>                                                  |
| (01/01/2027 to 31/12/2027)<br>Year 4 | <b>297,388.88</b>                                                  |
| (01/01/2028 to 31/12/2028)<br>Year 5 | <b>281,153.30</b>                                                  |

|                                        |                               |
|----------------------------------------|-------------------------------|
| (01/01/2029 to 31/12/2029)<br>Year 6   | <b>265,784.70</b>             |
| (01/01/2030 to 31/12/2030)<br>Year 7   | <b>251,237.50</b>             |
| (01/01/2031 to 31/12/2031)<br>Year 8   | <b>237,468.48</b>             |
| (01/01/2032 to 31/12/2032)<br>Year 9   | <b>224,436.69</b>             |
| (01/01/2033 to 31/12/2033)<br>Year 10  | <b>212,103.29</b>             |
| <b>Total estimated ERs</b>             | <b>2,428,045 (Round Down)</b> |
| <b>Total number of crediting years</b> | <b>10</b>                     |
| <b>Average annual ERs</b>              | <b>242,804 (Round Down)</b>   |

Approved By:



Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 30/04/2024

Place: Gurgaon, Haryana

# APPENDIX 1: DOCUMENTS REFERRED

| S.No. | Title of Document                                                                                                                                                                                       | Version                 | Date                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------|
| 1.    | Initial PD                                                                                                                                                                                              | 1.0                     | 01/01/2023                   |
| 2.    | Final PD                                                                                                                                                                                                | 2.5                     | 24/04/2024                   |
| 3.    | Estimated ER sheet                                                                                                                                                                                      | Pertaining to latest PD | -                            |
| 4.    | Applied methodology-<br><br>VMR0006 “Energy Efficiency and Fuel Switch Measures In Thermal Applications”<br><br>AMS-II.G. “Energy efficiency measures in thermal applications of non-renewable biomass” | V1.2<br><br>v13.0       | 06/07/2023<br><br>08/09/2022 |
| 5.    | VCS Project ID- 4416<br><br><a href="https://registry.verra.org/app/projectDetail/VCS/4416">https://registry.verra.org/app/projectDetail/VCS/4416</a>                                                   | -                       | -                            |
| 6.    | VCS Standard                                                                                                                                                                                            | 4.5                     | 04/10/2023                   |
| 7.    | VCS Program Guide                                                                                                                                                                                       | 4.4                     | 29/08/2023                   |
| 8.    | VCS Validation and verification manual                                                                                                                                                                  | 3.2                     | 19/10/2016                   |
| 9.    | Manufacturer’s Specification of G3 Burn rocket stove                                                                                                                                                    | -                       | 25/05/2022                   |
| 10.   | Baseline survey data sheets                                                                                                                                                                             | -                       | -                            |
| 11.   | Geographic location: google earth KML File                                                                                                                                                              | -                       | -                            |
| 12.1  | Declaration/ confirmation by PP that the project activity will not seek credits under any other program                                                                                                 | -                       | -                            |
| 12.2  | Declaration confirming that PP will own the Verified emission reductions                                                                                                                                | -                       | -                            |
| 12.3  | Declaration that the stoves are provided at zero cost                                                                                                                                                   | -                       | 05/05/2023                   |
| 13.   | Earthhood’s Remote Audit                                                                                                                                                                                | -                       | 22/09/2023                   |
| 14.   | WBT test report by CREEC Laboratory                                                                                                                                                                     | -                       | 25/05/2022                   |
| 15.   | Program Definitions                                                                                                                                                                                     | 4.4                     | 29/08/2023                   |
| 16.   | LSC Feedback forms                                                                                                                                                                                      | -                       | -                            |

|     |                                                                                                                                                                                                     |              |            |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| 17. | Email invitations, Newspaper advertisement and posters for LSC                                                                                                                                      | -            | -          |
| 18. | Attendance of LSC attendees, photos and material                                                                                                                                                    | -            | -          |
| 19. | LSC non-technical summary                                                                                                                                                                           | -            | -          |
| 20. | Tool 30: Calculation of the fraction of non-renewable biomass                                                                                                                                       | Version 4.0  | -          |
| 21. | Standard of sampling and survey of CDM Project activities and programme of activities                                                                                                               | Version 09.0 | -          |
| 22. | The Guidelines for sampling and survey for CDM Project activities and programme of activities                                                                                                       | Version 04.0 | -          |
| 23. | Pilot Stove distribution database                                                                                                                                                                   | -            | -          |
| 24. | Grievance Redress Policy and Procedure Manual,                                                                                                                                                      | Version 1.2  | -          |
| 25. | fNRB Calculation Sheet (TEM (2023)_fNRB calculation _FINAL.xlsx)                                                                                                                                    | -            | -          |
| 26. | TEM-APAC Security Review                                                                                                                                                                            | SecRisk      | 17/05/2023 |
| 27. | SPFEC Letter to TEM-Review of FnRB Calculations.pdf (third-party auditor)                                                                                                                           | -            | 17/03/2023 |
| 28. | End-user Training documents                                                                                                                                                                         | -            | -          |
| 29. | TEM Grievance Register                                                                                                                                                                              | -            | -          |
| 30. | End user beneficiary sample agreement                                                                                                                                                               | -            | -          |
| 31. | <ul style="list-style-type: none"> <li>TEM 2023 Cookstoves Highlands region Baseline survey report version 2</li> <li>SPFEC Letter_Cookstoves Highlands region baseline report version 2</li> </ul> | 2.0          | April 2024 |
| 32. | Registration and Issuance Process                                                                                                                                                                   | Version 4.3  | -          |
| 33. | Verra's Letter of Exemption for the on-site visit                                                                                                                                                   | -            | 30/08/2023 |
| 34. | Old UN Default values (EB67, Annex 22)                                                                                                                                                              | -            | -          |
| 35. | Hanson, L., Allen, B., Bourke, M., & McCarthy, T. (2001). Papua New Guinea Rural Development Handbook. (1st ed.) Australian National University.                                                    | -            | 2001       |
| 36. | Konedobu, P.N.G.: Energy Planning Unit, Department of Minerals and Energy, 1982                                                                                                                     |              |            |

|     |                                                                                                                                                                                                       |             |            |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 37. | ACIAR (Australian Centre for International Research). 2013. Promoting diverse firewood production systems in Papua New Guinea                                                                         | -           | 2013       |
| 38. | CDM Tool 33                                                                                                                                                                                           | Version 2.0 | -          |
| 39  | 2019 Refinement to 2006 IPCC Guidelines                                                                                                                                                               | -           | -          |
| 40. | Confirmation Letter by PNG Forest Authority                                                                                                                                                           | -           | -          |
| 41. | Page et al., (2016)                                                                                                                                                                                   | -           | 2016       |
| 42. | Murphy (2009) What is free about fuelwood? A critique of the value of fuelwood in the rural and squatter settlement households in the Eastern Highlands of Papua New Guinea                           | -           | 2009       |
| 43. | FAO (2020).Global Forest Resource Assessment, (FRA) PNG Report                                                                                                                                        | -           | 2020       |
| 44. | Papua New Guinea Forest Authority (2019)                                                                                                                                                              | -           | 2019       |
| 45. | 2011 National Population and Housing Census. Papua New Guinea 2011 National Report                                                                                                                    | -           | -          |
| 46. | IRENA (International Renewable Energy Agency), 2013. Report                                                                                                                                           | -           | -          |
| 47. | Signed end-user agreement for the cookstoves distributed during the pilot phase                                                                                                                       | -           | -          |
| 48. | a) TEM's end-user agreement for zero cost<br>b) TEM's funding letter for exclusion from government schemes and multilateral funds                                                                     | -           | May 2023   |
| 49. | GeoNarem Letter for fNRB_Jan 23.pdf (third-party forested that prepared the fNRB calculations)                                                                                                        | -           | -          |
| 50. | a) 4416_CDM_Sample Size Calculator_SRS-Mean[LIVE CALCULATOR].xlsx<br>b) 4416_CDM_Sample Size Calculation_SRS-Mean-Justification.pdf<br>c) 4416_PNG Cookstoves Sample Size Calculator [REFERENCE].xlsx | 1.0         | April 2024 |





## APPENDIX 2: ABBREVIATIONS

| Abbreviations   | Full texts                                            |
|-----------------|-------------------------------------------------------|
| BEF             | Baseline Emission Factor                              |
| CO <sub>2</sub> | Carbon dioxide                                        |
| CEA             | Central Electricity Authority                         |
| CERC            | Central Electricity Regulatory Commission             |
| CMS             | Central Monitoring System                             |
| CL              | Clarification Request                                 |
| CMP             | Conference of Parties Serving as Meeting of Parties   |
| CAR             | Corrective Action Request                             |
| EB              | Executive Board                                       |
| FAR             | Forward Action Request                                |
| GHG             | Green House Gas                                       |
| ICS             | Improved cookstoves                                   |
| MP              | Monitoring Period                                     |
| MR              | Monitoring Report                                     |
| PAI             | Project Activity Instances                            |
| PD              | Project Description                                   |
| PP              | Project Proponent                                     |
| QA/QC           | Quality Assurance and Quality Control                 |
| UNFCCC          | United Nations Framework Convention on Climate Change |
| VCS PD          | VCS Project Description                               |
| VCS             | Verified Carbon Standard                              |
| VCSA            | Verified Carbon Standard Association                  |
| VCUs            | Verified Carbon Units                                 |
| VVB             | Validation and Verification Body                      |

# APPENDIX 3: FINDINGS OVERVIEW

**Table 1. Remaining FAR from validation and/or previous verification**

|                                                      |    |                    |     |                          |
|------------------------------------------------------|----|--------------------|-----|--------------------------|
| <b>FAR ID</b>                                        | XX | <b>Section no.</b> | E.2 | <b>Date : DD/MM/YYYY</b> |
| <b>Description of FAR</b>                            |    |                    |     |                          |
|                                                      |    |                    |     |                          |
| <b>Project participant response</b>                  |    |                    |     | <b>Date : DD/MM/YYYY</b> |
|                                                      |    |                    |     |                          |
| <b>Documentation provided by project participant</b> |    |                    |     |                          |
|                                                      |    |                    |     |                          |
| <b>VVB assessment</b>                                |    |                    |     | <b>Date: DD/MM/YYYY</b>  |
|                                                      |    |                    |     |                          |

**Table 2. CL from this validation**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |                    |                   |                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|-------------------|--------------------------|
| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 01 | <b>Section no.</b> | Section 1.4 of PD | <b>Date : 15/05/2023</b> |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                    |                   |                          |
| <p>The PD mentions the lifetime of the ICS to be distributed as 10 years. However, the PP shall provide an evidential document substantiating the start date of the project activity.</p> <p>Also, the Section 1.4 mentions, “New Pis will be eligible for crediting from the start date of the instance through to the end of the project crediting period”, the PP shall substantiate the statement in context to the lifetime of the project device.</p>                                                                                                           |    |                    |                   |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |                    |                   | <b>Date : 17/05/2023</b> |
| <p>Statement has been substantiated with the following sentence:<br/> <i>For example, when a PI starts with a fixed crediting period of 10 years (fixed), the project device/stove must be proven at inclusion that it will be able to sustain through the crediting period.</i></p> <p>Document evidencing device lifespan will be sent to the validator.</p>                                                                                                                                                                                                        |    |                    |                   |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |                    |                   |                          |
| <ul style="list-style-type: none"> <li>- Manufacturer description of the project ICS including lifespan (file titled BURN G3 Stove Product Specifications.pdf)</li> <li>- Updated PD</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |    |                    |                   |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |                    |                   | <b>Date: 26/05/2023</b>  |
| <ul style="list-style-type: none"> <li>- The PP has provided a document ‘BURN G3 stove Product specification’, it has been verified from the manufacturers documents that the lifetime of the stove will be 10 years. Thus, it is acceptable.</li> <li>- The PP has decided to keep the crediting period for the 10 years (Fixed), and it has been also verified from the manufacturing specification that project lifetime is 10 years. Therefore, the explanation is found to be appropriate.</li> </ul> <p>The finding stands closed.</p> <p>CL# 01 is closed.</p> |    |                    |                   |                          |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|------------|--------------------------|
| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                   | 02 | <b>Section no.</b> | 1.11 of PD | <b>Date : 15/05/2023</b> |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                       |    |                    |            |                          |
| <p>Section 1.11 mentions the of Kuniokoa cookstove as the model of the stove to be distributed, however, the water boiling test submitted for the efficiency of the project stove mentions ‘G2 Burn Rocket stove’ as the model. (reference: BURN cookstove Water Boiling and Emission Tests.pdf). Please clarify the difference/ similarity in the stoves.</p> |    |                    |            |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                            |    |                    |            | <b>Date : 31/05/2023</b> |

The project device is marketed to end-users under the name Kuniokoa or Ecoa Wood Cookstove while its technical product name is the G3 Burn Rocket Stove, which is reflected in the WBT document. This wording has been included under Figure 1. In the PD for better clarity.

Also referencing the WBT document, the efficiency of the project device has been updated to 50.8% to reflect values stated in the submitted specification sheet.

As such, ER Calculations have also been updated to reflect the slight change in ERs.

#### Documentation provided by project participant

- Updated PD
- ER Calculation Sheet (Updated in Sharepoint)

#### VVB assessment

Date: 09/08/2023

The PP has satisfactorily explained the difference in the model naming of the product. Due to the company's rebranding efforts, the same product is being called G3 Burn Rocket Stove/ Kuniokoa or Ecoa Wood Cookstove. An email (from Burn) document provided by the PP also substantiates the same.

The VVB checked the updated PD and the ER Calculation sheet and found the changes to be in line with the above-given responses. Hence, the CL #02 stands CLOSED.

| CL ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 03 | Section no. | 3.2 of PD | Date | 15/05/2023        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|-----------|------|-------------------|
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |             |           |      |                   |
| <p>The section 3.2 mentions implementation in community-based Kitchen, however the baseline scenario mentioned is only for household. The PP shall explain the baseline scenario for community-based Kitchen in the PD.</p> <p>The section 3.2 mentions 'In order to avoid double counting of emission reductions through the method of distribution, each ICS will carry a unique identifier number'. The PP shall substantiate the procedure that will be applied for labelling the ICS to avoid double counting. Also, the PP shall explain how the PP will make sure that the stoves will not be double counted with any other project within the same geographical boundary.</p>                                                                                                                                                                                                                                                                                 |    |             |           |      |                   |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |             |           |      | Date : 17/05/2023 |
| <ol style="list-style-type: none"> <li>1) We amended the wording in the PD to only include households.</li> <li>2) Explanation for avoidance of double counting of ERs have been updated, see below:</li> </ol> <p>In order to avoid double counting of emission reductions through the method of distribution, each ICS will carry a Unique Identifier Number (UIN) and the distribution of each device is logged into an electronic database which includes unique information for each recipient (i.e., Unique Identification Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no UINs are double counted.</p> <p>During the monitoring period, survey samples will be only taken from the UINs that are in the electronic database of end-users eliminating the risk of double counting from any other projects within the same geographical location.</p> |    |             |           |      |                   |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |             |           |      |                   |
| <ul style="list-style-type: none"> <li>- Updated PD</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |             |           |      |                   |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |             |           |      | Date: 26/05/2023  |
| <p>The PP has revised the section and has mentioned only households. It has been verified that the stoves will only be distributed in the households not in commercial- based kitchen.</p> <p>PP has updated the section and it has been confirmed that each stove distributed will be identified using the UIN provided to it. A data base/ record will also be maintained to avoid any kind of double counting.</p> <p>Therefore, the finding CL#03 stands closed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |             |           |      |                   |

| CL ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 04 | Section no. | 4.1 of PD | Date : 15/05/2023        |
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| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |             |           |                          |
| After the distribution of project devices to the end-users, the baseline device (conventional three-stone stove) may still be used by some of the end-users. PP is requested to clarify the following:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |             |           |                          |
| <ol style="list-style-type: none"> <li>1. What are the measures taken for the discontinued use of traditional stove.</li> <li>2. In case of any continued use of traditional stove, how the factor will be taken into emission reduction calculation.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |             |           |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |             |           | <b>Date : 17/05/2023</b> |
| <ol style="list-style-type: none"> <li>1) Currently, Tasman Environmental Markets Asia Pacific Pte. Ltd doesn't have measures in place for the discontinued traditional stove. Instead, Tasman Environmental Markets Asia Pacific Pte. Ltd intends to encourage the continued use and uptake of the project ICS throughout the implementation of the project.</li> <li>2) In the event traditional cookstoves are still being used, Tasman Environmental Markets Asia Pacific Pte. Ltd will be monitoring the fraction of traditional cookstoves still being used during the crediting period through the <math>\mu_y</math> parameter, which adjusts the <math>B_{old,i}</math> parameter for a more accurate baseline measure on firewood consumed in the baseline. Monitoring will likely occur within the first year of ICS installation.</li> </ol> |    |             |           |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |             |           |                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |             |           |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |             |           | <b>Date: 26/05/2023</b>  |
| The PP has mentioned the parameters in the registered monitoring plan, the continued use of baseline stove will be monitored using parameter $\mu_y$ . the information has been cross verified from the monitoring plan. Therefore, the finding stands closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |             |           |                          |
| CL#04 Closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |             |           |                          |

| CL ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 05 | Section no. | 4 of PD | Date : 15/05/2023        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|---------|--------------------------|
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |             |         |                          |
| PP is requested to clarify how the consumption of wood in accordance with the seasonal variation will be accounted for. Request to update the VCS PD accordingly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |             |         |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |             |         | <b>Date : 17/05/2023</b> |
| <p>Regarding the quantity of biomass consumed by project ICS (tonnes per household per year) known as <math>B_{old,i}</math>, the project has estimated this parameter during the baseline study. There is unlikely this value will be subject to seasonal fluctuation (ACIAR, 2013).</p> <p>However, to bring some additional accuracy to the proposed value, the project will monitor this parameter during the first year of ICS installation through the use of visual inspections, questionnaires and firewood measurements following the pre-established baseline protocols. Then, this parameter will be fixed for the entire crediting period.</p> <p>This reasoning is also justified by the following source in Section 5.1:<br/>ACIAR (Australian Centre for International Research). 2013. Promoting diverse firewood production systems in Papua New Guinea</p> |    |             |         |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |             |         |                          |
| - Updated PD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |             |         |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |             |         | <b>Date: 26/05/2023</b>  |
| The PP has provided the appropriate and the information from the sources provided by the PP is also cross checked. Therefore, the finding stands closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |             |         |                          |
| CL #05 closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |             |         |                          |

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| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 06 | <b>Section no.</b> | 1.4 of PD | <b>Date</b> | 26/05/2023   |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |                    |           |             |              |
| In accordance with the document 'BURN G3 stove product specification' the warranty of the product is 1 year. The PP shall explain what will be done if the stove is damaged or it breaks after 1 year. Will people have to buy stoves?                                                                                                                                                                                                                           |    |                    |           |             |              |
| In accordance with the information mentioned, the project activity has 2 types of stoves (with and without adjustable skirt). The PP shall clarify the basis of deciding the type of stove that will be distributed in different households.                                                                                                                                                                                                                     |    |                    |           |             |              |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                    |           |             | <b>Date</b>  |
| There are no measures that will be taken if the stove is damaged or breaks after 1 year. Instead, Tasman Environmental Markets Asia Pacific Pte. Ltd intends to train the community and end-users on how to best maintain the project device throughout the implementation of the project. This may be done through flyers that may be given out together with the project stove during implementation and maintenance checks done biannually by the local team. |    |                    |           |             | 31/05/2023   |
| To clarify, the project will only distribute one type of stove, the G3 Burn Rocket Stove with adjustable pot skirt. This has been clarified in Section 1.11 of the PD where Figure 1 only shows one cookstove.                                                                                                                                                                                                                                                   |    |                    |           |             |              |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                             |    |                    |           |             |              |
| - Updated PD                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                    |           |             |              |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |                    |           |             | <b>Date:</b> |
| The information provided by the PP is found to be sufficient to close the finding. PP has updated the PD to reflect only one type of stove, the G3 Burn Rocket Stove with adjustable pot skirt and is found to be appropriate. Hence, the finding is now closed.                                                                                                                                                                                                 |    |                    |           |             | 26/05/2023   |
| CL#06 closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                    |           |             |              |

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| <b>CL ID</b>                                                                                                                                                                                                                                                                                     | 07 | <b>Section no.</b> | 3.2 of PD | <b>Date</b> | 26/05/2023   |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                         |    |                    |           |             |              |
| PP shall provide a declaration/ contract between the stove manufactures/ wholesalers regarding the transfer of carbon rights or they will not claim any carbon reductions resulting from the project activity.                                                                                   |    |                    |           |             |              |
| <b>Project participant response</b>                                                                                                                                                                                                                                                              |    |                    |           |             | <b>Date</b>  |
| With reference to the document 'Burn Invoice_TEM_AUG_2022' within the Share point, the project stoves are sold on a one-off basis whereby the manufacturer sells the project devices in its entirety (all rights attached to the product) to Tasman Environmental Markets Asia Pacific Pte. Ltd. |    |                    |           |             | 31/05/2023   |
| There are no additional caveats attached to the purchase of the project stoves.                                                                                                                                                                                                                  |    |                    |           |             |              |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                             |    |                    |           |             |              |
| - Burn Invoice_TEM_AUG_2022.pdf                                                                                                                                                                                                                                                                  |    |                    |           |             |              |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                            |    |                    |           |             | <b>Date:</b> |
| The referred document was reviewed by the assessment team and was found to be acceptable. Hence, the finding is CLOSED.                                                                                                                                                                          |    |                    |           |             | 09/08/2023   |

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| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                     | 08 | <b>Section no.</b> | 3.2 of PD | <b>Date</b> | 09/08/2023  |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                         |    |                    |           |             |             |
| The applied methodology VMR0006 mandates the application of the CDM methodology AMS-II.G v13.0 for the project activity. Section 3.2, applicability of AMS-II.G v13.0, point 2, mentions 'Any additional emission savings will be counted within a new project instance'. PP is requested to clearly define the project instances in the VCS PD. |    |                    |           |             |             |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                              |    |                    |           |             | <b>Date</b> |
|                                                                                                                                                                                                                                                                                                                                                  |    |                    |           |             | 10/08/2023  |

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| Included additional language to state the following:<br>Each project instance will consist of no more than 100,000 ICSSs. In the event a Project Instance exceeds 100,000 ICSSs, a new Project Instance must be included.         |                         |
| <b>Documentation provided by project participant</b>                                                                                                                                                                              |                         |
| - Updated PD                                                                                                                                                                                                                      |                         |
| <b>VVB assessment</b>                                                                                                                                                                                                             | <b>Date: 16/08/2023</b> |
| The PP has updated the PD to clearly specify one 'project instance' as the distribution of 100,000 ICSSs, which is found to be appropriate as per VCS Standard, hence, acceptable. The assessment team accepts the clarification. |                         |
| Thus, CL#08 is CLOSED.                                                                                                                                                                                                            |                         |

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| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 09 | <b>Section no.</b> | 3.4 of PD | <b>Date : 26/05/2023</b> |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |                    |           |                          |
| It has been mentioned that the baseline scenario of the project activity is "continued use of non-renewable wood fuel (firewood) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cook stoves in absence of project activity". PP is requested to clarify how was the baseline scenario and prevalence of traditional stoves established? PP is requested to clarify what method is used for the determination of baseline scenario and its appropriateness. Request to update the section 3.4 of VCS PD accordingly. |    |                    |           |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |                    |           | <b>Date : 11/08/2023</b> |
| A third-party consultant team contracted by Tasman Environmental Markets Asia Pacific Pte. Ltd conducted a baseline survey during October 2022 with the results included in a report that was audited by an esteem in country forester (also included in the evidence provided).<br>File Reference: <a href="#">Cookstoves Highlands region report March 2023.pdf</a>                                                                                                                                                                                                                    |    |                    |           |                          |
| Findings of the baseline survey has been included into the Section 3.4 of the Section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |                    |           |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                    |           |                          |
| - Updated PD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |                    |           |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                    |           | <b>Date: 16/08/2023</b>  |
| The above-referred report was checked, and the PP has clarified the process of determination of the baseline scenario in the region. It is now clear that in baseline the major population in Papua New Guinea use firewood along with the use of traditional cookstoves, which is further supported by several studies in the report (like ACIAR, 2013, IRENA report of 2013 and Page et al. (2016)). However, PP is requested to elaborate how the use of traditional cookstoves is taken as a baseline scenario.<br>The finding remains partially OPEN.                               |    |                    |           |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |                    |           | <b>Date : 04/09/2023</b> |
| Additional information on the Baseline Survey's findings with regards to PNG's use of traditional cooking methods (Open Drum Stoves and Open Fire Cooking) has been added into the PD.                                                                                                                                                                                                                                                                                                                                                                                                   |    |                    |           |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                    |           |                          |
| - Updated PD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |                    |           |                          |
| <b>VVB Assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                    |           | <b>Date : 27/09/2023</b> |
| The PD has been updated to sufficiently elaborate the baseline scenario where the use of Open Drum Stoves and Open Fire Cooking is rampant. The explanation is assessed and found to be appropriate.<br><br>Hence the finding CL#09 is CLOSED.                                                                                                                                                                                                                                                                                                                                           |    |                    |           |                          |

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| <b>CL ID</b>             | 10 | <b>Section no.</b> |  | <b>Date : 26/05/2023</b> |
| <b>Description of CL</b> |    |                    |  |                          |

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| As per section 8.3 of VMR0006, "Projects must apply an adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (Adj <sub>LE</sub> ). The procedures from the latest version of AMS-II.G. must be applied. Adj <sub>LE</sub> is fixed at validation." However, in section 4.3 of VCS PD it has been mentioned that PP will be applying the adjustment factor for f <sub>NRB</sub> in its calculations, the adjustment factor Adj <sub>LE</sub> will not be applied. PP is requested to clarify how the adjustment factor has been applied in line with the procedures and requirements in AMS-II.G and VMR0006. |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Date : 11/08/2023</b> |
| Leakage for the project activity has been amended to take into consideration leakage related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced.                                                                                                                                                                                                                                                                                                                                                                                                |                          |
| Project participant will take Adj <sub>LE</sub> as 0.95, therefore sampling is not required. This number will be fixed at validation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |
| <ul style="list-style-type: none"> <li>- Updated PD</li> <li>- Updated ER Calculations Sheet</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Date: 16/08/2023</b>  |
| The PP has updated the PD to add the fact that Adj <sub>LE</sub> is fixed at 0.95. The same has been reflected in the ER sheet also, which has resulted in a slight decline in ERs. The assessment team found the changes in line with the applied version of the VMR0006 and AMS-II.G methodology.                                                                                                                                                                                                                                                                                                                                                                      |                          |
| Thus, the finding is CLOSED.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |

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| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 11 | <b>Section no.</b> | RSV | <b>Date : 28/09/2023</b> |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |                    |     |                          |
| During the remote audit of the Western Highlands Provinces (WHP), it was observed that a few recipients (under the pilot-run) owned and used gas stoves and electric stoves, in addition to the ICS. The PP is requested to clarify upon the claimed identified baseline and the recipient selection criteria for the distribution of ICS under the current project.                                                                                                                                                                                                                                                                                           |    |                    |     |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |                    |     | <b>Date : 29/09/2023</b> |
| A third-party consultant team contracted by Tasman Environmental Markets Asia Pacific Pte. Ltd conducted a baseline survey during October 2022 with the results included in a report that was audited by an esteem in country forester (also included in the evidence provided). File Reference: <u>Cookstoves Highlands region report March 2023.pdf</u> . As established by VCS the purpose of the baseline is to gather information on the current cooking practices in these local communities to understand their cooking behaviours and practices. This, in turn, will enable a better understanding of the additional of the proposed project activity. |    |                    |     |                          |
| The results from the baseline survey showed that 98% of the houses involved in the study used firewood as a primary source of energy, whereas only 2% of total households located in semi-urban settlements mentioned the use of electricity or gas as secondary energy sources (p. 25).                                                                                                                                                                                                                                                                                                                                                                       |    |                    |     |                          |
| The inclusion of these houses in our baseline may have slightly reduced our B <sub>old</sub> estimation on firewood consumption/house/year. However, we have decided to take a conservative approach without foreseen any changes in the proposed estimates.                                                                                                                                                                                                                                                                                                                                                                                                   |    |                    |     |                          |
| We will also clearly state in the PD (p 04) that only rural households that use firewood as a main source of energy will be involved in the project and receive an improved cookstove (ICS). To ensure this, the consent form end-users are required to sign in order to receive their ICS in the <u>full-scale project</u> will include a self-declaration indicating that firewood is their primary source of cooking fuel.                                                                                                                                                                                                                                  |    |                    |     |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |                    |     |                          |
| <ul style="list-style-type: none"> <li>- Updated PD</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |                    |     |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |                    |     | <b>Date: 03/10/2023</b>  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| The assessment team checked the baseline survey which indicated that 98% of the households use firewood for cooking purposes. But during the RSV and through end-user consent form, it was observed that 2 out of the 6 end-users in the Western Highlands Region used gas stove and electric stove for cooking. The PP is, therefore, requested to clarify the criteria for selecting the households for ICS distribution and how is it being ensured that the households using open-fire and open drum-stove for cooking are specifically targeted.<br>The finding CL#11 stands open.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Date : 05/10/2023</b> |
| Tasman Environmental Markets Asia Pacific Pte. Ltd will apply a “household selection criteria” during the process of selecting houses. This selection criteria will serve as a confirmation that each house is aligned with the baseline scenario set out for the Project activity.<br><br>The selection criteria will take the form of a questionnaire at the point of ICS distribution, and will include binomial questions (yes/no answers) to access the household’s primary method of cooking fuel as follows:<br><ol style="list-style-type: none"> <li>1. Do you cook any meals in the house without using firewood?</li> <li>2. Do you use charcoal for cooking in your household?</li> <li>3. Do you use electric cooking in your household?</li> <li>4. Do you use LPG for cooking in your household?</li> </ol> If any of the responses to these questions indicates that firewood is not the main fuel used in the household for cooking, the household will not be selected to participate in the proposed project activity. |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |
| - Updated PD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Date : 18/10/2023</b> |
| The information provided by the PP has been reviewed and the selection criteria that will be implemented is found to be appropriate.<br><br>Hence, CL#11 is CLOSED.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                    |       |                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|-------|--------------------------|
| <b>CL ID</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12 | <b>Section no.</b> | 3.3.8 | <b>Date : 18/10/2023</b> |
| <b>Description of CL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |                    |       |                          |
| Following observations has been made during the TR Round:<br><ol style="list-style-type: none"> <li>1. It has been observed through the baseline survey report that CCT has been conducted by the PP, PP is requested to clarify why equation 9 of the AMS-II.G (para 33) has not been followed to calculate the emission reduction calculation. PP is requested to revise the relevant section (if applicable) accordingly.</li> <li>2. There are 4 provinces for ICS distribution, however only 2 have been considered for baseline survey. The PP shall clarify the appropriateness of the baseline survey being conducted in only 2 province, whereas the project is estimated to be implemented in 4 province of PNG.</li> <li>3. It has been observed that Parameter <math>\eta_{new,i,j}</math> has been added in section 5.1 ‘Data and parameter available at validation’, which is found to be inconsistent with the AMS-II.G, where the parameter is added under section 6.2, please clarify. <ol style="list-style-type: none"> <li>1) Additionally, PP is requested to clarify how the loss of efficiency will be determined as per para 37 of AMS-II.G throughout the lifetime of the project.</li> </ol> </li> </ol> |    |                    |       |                          |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |                    |       | <b>Date : 19/10/2023</b> |
| 1. It has been observed through the baseline survey report that CCT has been conducted by the PP, PP is requested to clarify why equation 9 of the AMS-II.G (para 33) has not been followed to calculate the emission reduction calculation. PP is requested to revise the relevant section (if applicable) accordingly.<br>Response:<br><b>Reasoning behind selection of Equation (7) of AMS-II.G:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |                    |       |                          |



At project inception, the requirements state the need for baseline data that can be extracted from published peer-review scientific articles or through the use of survey-based questionnaires to collect data in the proposed project area. At this stage control cooking tests (CCTs) are not compulsory.

As per the report submitted, the baseline study was conducted during October 2022 and it was aligned with the specifications and requirements set out in VMR0006 Version 1.1.

VMR0006 v1.1 (page 10, footnote 10) states the option to determine  $B_{y,savings}$  prior to the validation of the project.  $B_{y,savings}$  was determined using Equation 3, using the parameters  $B_{old}$ ,  $\eta_{new,i,j}$ , (supported by the water boiling test report from the manufacturer) and  $\eta_{old}$ . This is reflected in the first ER Calculations sheet submitted to the VVB (5th May 2023)

This equation is still consistent with VMR0006 v. 1.2, which references paragraphs 32 and Equation (7) from AMS-II.G v13 for the  $B_{y,savings}$  to be estimated with a water boiling test (WBT).

**CCT conducted by Project Participant (Tasman Environmental Markets Asia Pacific Pte. Ltd):**

Tasman Environmental Markets Asia Pacific Pte. Ltd is currently developing the first improved cookstove project in Papua New Guinea (PNG) where robust scientific data is scarce. The original design of the control cooking tests (CCTs) aimed to understand the differences in firewood consumption between Burn cookstoves (ICS) and current open fire practices in PNG. As Tasman Environmental Markets Asia Pacific Pte. Ltd initially decided to use Equation 3 of VMR0006 v1.1, there was no compulsory need for CCTs. As such, data derived from these tests serve as a good proxy to understand the degree of acceptance of this new technology across the local communities, the main co-benefits of the project (e.g., Time saved during cooking, reduction of smoke produced, savings in firewood used, etc.), and none of the technical parameters required under AMS-II.G v13.0 Equation 9 were collected.

**No values obtained from the CCT have been incorporated into the ER Calculations.**

The  $B_{old}$  value of 5.56 tonnes/year/ household that Tasman Environmental Markets Asia Pacific Pte. Ltd is using in its ER Calculations was extracted from the survey questionnaires considering the preparation of 2 meals per day and is presented in Table 2 of the Baseline report (File titled: Cookstoves Highlands region report March 2023.pdf). This estimate is more conservative than the firewood consumption value of 8.77 tonnes/year/household estimated from the control cooking tests considering 2 meals per day.

2. There are 4 provinces for ICS distribution, however only 2 have been considered for baseline survey. The PP shall clarify the appropriateness of the baseline survey being conducted in only 2 province, whereas the project is estimated to be implemented in 4 province of PNG.

Response:

As mentioned in the CL ID 9 and 11, a third-party consultant team contracted by Tasman Environmental Markets Asia Pacific Pte. Ltd conducted a baseline survey during October 2022 with the results included in a report that was audited by an esteemed country forester (also included in the evidence provided). File titled: Cookstoves Highlands region report March 2023.pdf

In this peer-reviewed report, the results of the baseline study were placed under the broader context of Highlands region, based on the following reasons:

- 1) The available literature shows that the four neighbouring provinces proposed for the project activity (Southern Highlands Province, Western Highlands Province, Jiwaka Province, and Eastern Highlands Province) are characterized by similar geography that is dominated by high mountains (over 1,000 mts above sea level), that lead to the lowest average temperature in

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <p>the country are within a similar socio-economic and ecological conditions, that lead to similar behaviours in the dependence of firewood as a main energy source for cooking (Hanson et al., 2001, ACIAR, 2013).</p> <ol style="list-style-type: none"> <li>2) There are 3 published peer-reviewed research studies on firewood consumption for cooking and heating that collected data in different provinces in the Highlands region. Murphy (2009) estimated the average firewood consumption as 3.64 tonnes/household/year, ACIAR (2013) proposed that 7.51 tonnes/household/year and Page et al., (2016) estimated an average of 11.10 tonnes/household/year.</li> <li>3) While the range of values extracted from the literature (3.64-11.10 tonnes/household/year) include research in different provinces in the Highlands region, it is likely that these estimates of firewood consumption are rather related to general practices than an accurate representation of the value attributed to cooking practices. Our baseline study showed that the estimated firewood consumption for cooking was 5.56 tonnes/household/year and provides a representative conservative estimate of the firewood consumption for cooking across the Highlands region (including the Southern Highlands Province, Western Highlands Province, Jiwaka Province and Eastern Highlands Province).</li> </ol> <p>Based on the information presented in points 1) and 2), and in addition to well-known security risks in the area, with the third-party consultant technical team (Simi Oa Consultancy services) we have decided to collect information in two provinces that will be representative of the firewood consumption behaviours across the Highlands region.</p>                                                                                                                                                                                   |                                |
| <p>3. It has been observed that Parameter <math>\eta_{new,i,j}</math> has been added in section 5.1 'Data and parameter available at validation', which is found to be inconsistent with the AMS-II.G, where the parameter is added under section 6.2, please clarify.<br/>Additionally, PP is requested to clarify how the loss of efficiency will be determined as per para 37 of AMS-II.G throughout the lifetime of the project.</p> <p>Parameter <math>\eta_{new,i,j}</math> has been moved to Section 5.2 of the PD, Data and Parameters Monitored, to maintain consistency with AMS-II.G. The parameter will also be included in the sampling plan as a parameter to be sampled and monitored.</p> <p>Loss in efficiency of the project devices due to aging during the monitoring period will be accounted for using the following methods set out in AMS-II.G v13.0:</p> <ol style="list-style-type: none"> <li>1. Manufacturer of project devices shall confirm with technical justification based on certification by a national standards body or an appropriate certifying agent recognized by that body that no decrease in efficiency of project device is envisaged during the crediting period; or</li> <li>2. Determine the rate of efficiency drop for a representative sample of the first batch of project device <math>i</math> in year <math>y</math> and assume that same rate of loss in efficiency applies to all other batches. In other words, it may be assumed that the degradation of efficiency measured in a representative sample of the first batch of project devices <math>i</math> apply to all subsequent batches. The efficiency of the project devices in the first batch has to be monitored annually through representative samples and this rate of loss in efficiency may be applied correspondingly to all batches.</li> </ol> <p>In the event that option 1 is not available, option 2 will be used.</p> |                                |
| <p><b>Documentation provided by project participant</b></p> <ul style="list-style-type: none"> <li>- Updated PD</li> <li>- PNG Cookstoves LSC Participant List</li> <li>- PNG Cookstoves LSC Report</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                |
| <p><b>DOE assessment</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>Date:</b> 20/10/2023</p> |

1. The information provided by the PP has been reviewed and it now clear that at the time when baseline survey was conducted (October 2022), the applicable methodology was VMR0006 v1.1 and all the survey questionnaire has been prepared in line with the applicable version of methodology and the value of the parameter has been derived on the basis of the same. At a later stage, CCT was also performed by the PP, to understand the requirements of the locals and the efficiency and performance of the distributed stoves. It has also been observed that the value applied by the PP is more conservation than the value obtained by through the CCT and hence, the approach applied by the PP is found to be appropriate. CLOSED
2. The PP has appropriately demonstrated the rationale behind conducting the baseline survey in two of the four provinces where the ICSs are to be distributed. The documents referred to by the PP have also been assessed by the DOE team and found to be appropriate. CLOSED
3. The PP has added the said parameter under the appropriate section, i.e., 6.2 'Data and Parameters Monitored.' Additionally, the loss of efficiency has been factored in, and during the monitoring, it will be calculated as per one of the two options chosen by the PP (out of the specified options from the methodology AMS-II.G v13.0. The assessment team found the response to be appropriate. CLOSED

Hence, the finding CL#12 stands CLOSED.

**Table 3. CAR from this validation**

| CAR ID                                                                                                                                                                   | 01 | Section no. |  | Date : 15/05/2023        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|--|--------------------------|
| <b>Description of CAR</b>                                                                                                                                                |    |             |  |                          |
| The PP shall mention the start date of the group project and the crediting period in the PD, wherever applicable.                                                        |    |             |  |                          |
| <b>Project participant response</b>                                                                                                                                      |    |             |  | <b>Date : 17/05/2023</b> |
| Start date of GPA has been updated to: 01/12/2023.                                                                                                                       |    |             |  |                          |
| Crediting period has been updated to: 01/12/2023 to 31/11/2032                                                                                                           |    |             |  |                          |
| <b>Documentation provided by project participant</b>                                                                                                                     |    |             |  |                          |
| Updated PD                                                                                                                                                               |    |             |  |                          |
| <b>VVB assessment</b>                                                                                                                                                    |    |             |  | <b>Date: 26/05/2023</b>  |
| The PP has updated the PD, start date of the project and start date of crediting period have been mentioned in the relevant sections of the PD.<br>CAR 01 stands Closed. |    |             |  |                          |

| CAR ID                                                                                                                                                                                                      | 02 | Section no. | PD | Date : 15/05/2023        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|----|--------------------------|
| <b>Description of CAR</b>                                                                                                                                                                                   |    |             |    |                          |
| Minor Corrections required-<br>- In section 3.2, PP the shall mention the scale of the project.<br>- In section 2.4, PP shall mention the duration of public comment period.                                |    |             |    |                          |
| <b>Project participant response</b>                                                                                                                                                                         |    |             |    | <b>Date : 17/05/2023</b> |
| 1) The scale of the project activity has been clarified to be 'Large-Scale' in Section 3.2 of the PD.<br>2) Duration of public comment period has been included in Section 2.4, Public Comments.            |    |             |    |                          |
| <b>Documentation provided by project participant</b>                                                                                                                                                        |    |             |    |                          |
| - Updated PD                                                                                                                                                                                                |    |             |    |                          |
| <b>VVB assessment</b>                                                                                                                                                                                       |    |             |    | <b>Date: 26/05/2023</b>  |
| The PP has revised section 3.2 and section 2.4, mentioning the scale of the project and duration of public comment period respectively. The information is found to be appropriate.<br><br>CAR # 02 closed. |    |             |    |                          |

|                                                                                                                                |    |                    |    |              |            |
|--------------------------------------------------------------------------------------------------------------------------------|----|--------------------|----|--------------|------------|
| <b>CAR ID</b>                                                                                                                  | 03 | <b>Section no.</b> | PD | <b>Date</b>  | 26/05/2023 |
| <b>Description of CAR</b>                                                                                                      |    |                    |    |              |            |
| PP shall update the template to the latest available template on VERRA WEBSITE, version 4.2                                    |    |                    |    |              |            |
| <b>Project participant response</b>                                                                                            |    |                    |    | <b>Date</b>  | 31/05/2023 |
| The PD has been updated to the latest version according to the Verra Website.                                                  |    |                    |    |              |            |
| <b>Documentation provided by project participant</b>                                                                           |    |                    |    |              |            |
| - Updated PD                                                                                                                   |    |                    |    |              |            |
| <b>VVB assessment</b>                                                                                                          |    |                    |    | <b>Date:</b> | 09/08/2023 |
| The PP has updated the PD template as per version 4.2 available on the website of Verra. Hence, the finding CAR #03 is CLOSED. |    |                    |    |              |            |

|                                                                                                                                             |    |                    |                       |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|-----------------------|--------------|------------|
| <b>CAR ID</b>                                                                                                                               | 04 | <b>Section no.</b> | ER Calculations Sheet | <b>Date</b>  | 09/08/2023 |
| <b>Description of CAR</b>                                                                                                                   |    |                    |                       |              |            |
| In the ER sheet provided by the PP, cell Q7 says $\mu_d$ , while it should be $(1-\mu_d)$ . The PP shall rectify the error.                 |    |                    |                       |              |            |
| <b>Project participant response</b>                                                                                                         |    |                    |                       | <b>Date</b>  | 10/08/2023 |
| This has been updated in the ER Calculations sheet.                                                                                         |    |                    |                       |              |            |
| <b>Documentation provided by project participant</b>                                                                                        |    |                    |                       |              |            |
| - Updated ER Calculation Sheet                                                                                                              |    |                    |                       |              |            |
| <b>VVB assessment</b>                                                                                                                       |    |                    |                       | <b>Date:</b> | 16/08/2023 |
| The error has been corrected in the ER sheet and the abovesaid cell reads $(1-\mu_d)$ instead of $\mu_d$ . Hence the finding stands CLOSED. |    |                    |                       |              |            |

|                                                                                                                                                                                                          |    |                    |             |              |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|-------------|--------------|------------|
| <b>CAR ID</b>                                                                                                                                                                                            | 05 | <b>Section no.</b> | 5 of VCS PD | <b>Date</b>  | 09/08/2023 |
| <b>Description of CAR</b>                                                                                                                                                                                |    |                    |             |              |            |
| Parameters 'ADJLE' and 'ud' are not mentioned in section 5 of VCS PD in line with section 9.1 of the applied methodology VMR0006 v1.2. PP is requested to take corrective action.                        |    |                    |             |              |            |
| <b>Project participant response</b>                                                                                                                                                                      |    |                    |             | <b>Date</b>  | 11/08/2023 |
| Included into validation parameters.                                                                                                                                                                     |    |                    |             |              |            |
| <b>Documentation provided by project participant</b>                                                                                                                                                     |    |                    |             |              |            |
| - Updated PD                                                                                                                                                                                             |    |                    |             |              |            |
| <b>VVB assessment</b>                                                                                                                                                                                    |    |                    |             | <b>Date:</b> | 16/08/2023 |
| The PD has been updated accordingly, to reflect both the parameters as per the methodology under consideration. The information in the parameters has also been assessed and is found to be appropriate. |    |                    |             |              |            |
| Thus, the finding is CLOSED.                                                                                                                                                                             |    |                    |             |              |            |

**Table 4. FAR from this validation**

|                                                                                                                                                                                                                                                                                                                                                   |    |                    |    |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------|----|--------------|------------|
| <b>FAR ID</b>                                                                                                                                                                                                                                                                                                                                     | 01 | <b>Section No.</b> | NA | <b>Date</b>  | 30/10/2023 |
| <b>Description of FAR</b>                                                                                                                                                                                                                                                                                                                         |    |                    |    |              |            |
| In Line with Section 4.1.12 of the VCS Standard v4.5, during the first verification of the project activity, the next verifying VVB shall conduct an on-site visit of the project activity to reassess the following information provided by the project proponent: Project Location, Baseline Scenario, Stakeholder Engagement and consultation. |    |                    |    |              |            |
| <b>Project participant response</b>                                                                                                                                                                                                                                                                                                               |    |                    |    | <b>Date</b>  | DD/MM/YYYY |
|                                                                                                                                                                                                                                                                                                                                                   |    |                    |    |              |            |
| <b>Documentation provided by project participant</b>                                                                                                                                                                                                                                                                                              |    |                    |    |              |            |
|                                                                                                                                                                                                                                                                                                                                                   |    |                    |    |              |            |
| <b>VVB assessment</b>                                                                                                                                                                                                                                                                                                                             |    |                    |    | <b>Date:</b> | DD/MM/YYYY |
|                                                                                                                                                                                                                                                                                                                                                   |    |                    |    |              |            |



# APPENDIX 4: COMPETENCY STATEMENTS

| Competence Statement |                                                                    |      |            |
|----------------------|--------------------------------------------------------------------|------|------------|
| Name                 | Divij Varshney                                                     |      |            |
| Education            | M.Tech. Renewable energy systems<br>B.Tech. Electrical Engineering |      |            |
| Experience           | 1.5 years                                                          |      |            |
| Field                | Climate Change & Environment / Industry                            |      |            |
| Approved Roles       |                                                                    |      |            |
| Team Leader          | Yes (VM)                                                           |      |            |
| Validator            | Yes (VM)                                                           |      |            |
| Verifier             | Yes (VM)                                                           |      |            |
| Local expert         | Yes (India)                                                        |      |            |
| Financial Expert     | NO                                                                 |      |            |
| Technical Reviewer   | NO                                                                 |      |            |
| TA Expert (X.X)      | Yes (VM TA 1.2, 3.1)                                               |      |            |
|                      |                                                                    |      |            |
| Reviewed by          | Shifali Guleria, Quality Manager                                   | Date | 16/04/2023 |
| Approved by          | Deepika Mahala, Technical Manager                                  | Date | 16/04/2023 |

| Competence Statement |                                                                                                            |      |            |
|----------------------|------------------------------------------------------------------------------------------------------------|------|------------|
| Name                 | Deepika Mahala                                                                                             |      |            |
| Education            | M. Sc. (Environment Management), GGSIP University<br>B.Sc. Hons. (Chemistry), Sri Venkateshwar College, DU |      |            |
| Experience           | 6 Years +                                                                                                  |      |            |
| Field                | Climate Change                                                                                             |      |            |
| Approved Roles       |                                                                                                            |      |            |
| Team Leader          | YES                                                                                                        |      |            |
| Validator            | YES                                                                                                        |      |            |
| Verifier             | YES                                                                                                        |      |            |
| Local expert         | YES (India, Bangladesh)                                                                                    |      |            |
| Financial Expert     | NO                                                                                                         |      |            |
| Technical Reviewer   | YES                                                                                                        |      |            |
| TA Expert            | YES (TA 1.2 & TA 3.1)<br>(TA 1.1, 13.1, 13.2 – VM only)                                                    |      |            |
|                      |                                                                                                            |      |            |
| Reviewed by          | Shifali Guleria (QM)                                                                                       | Date | 28/06/2023 |
| Approved by          | Kaviraj Singh (MD)                                                                                         | Date | 28/06/2023 |

| Competence Statement |                                         |      |            |
|----------------------|-----------------------------------------|------|------------|
| Name                 | Vaishali Yadav                          |      |            |
| Education            | M.Sc. environmental studies             |      |            |
| Experience           | -                                       |      |            |
| Field                | Climate Change & Environment / Industry |      |            |
| Approved Roles       |                                         |      |            |
| Team Leader          | NO                                      |      |            |
| Validator            | NO                                      |      |            |
| Verifier             | NO                                      |      |            |
| Local expert         | NO                                      |      |            |
| Financial Expert     | NO                                      |      |            |
| Technical Reviewer   | NO                                      |      |            |
| TA Expert (X.X)      | NO                                      |      |            |
| Trainee              | YES                                     |      |            |
|                      |                                         |      |            |
| Reviewed by          | Shifali Guleria (Quality Manager)       | Date | 12/07/2023 |
| Approved by          | Deepika Mahala (Technical Manager)      | Date | 12/07/2023 |